

8

MEASUREMENT AND SPACE

FURTHER TRIGONOMETRY

Trigonometry is a branch of mathematics that uses the relationship between the angles and sides of triangles. Trigonometry is used in navigation when locating the position or bearing of a destination from a known location, and finding the distances between places that cannot be physically measured.

Trigonometry helps us understand physical phenomena that are periodic or cyclic, such as tidal movements, phases of the moon, average monthly temperatures and sound waves. It is also used in construction, engineering, physics and gaming.



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Chapter outline

		Working mathematically				
8.01	Right-angled trigonometry	U	F	PS		C
8.02	Angles of elevation and depression	U	F	PS		
8.03	Bearings	U	F	PS	R	C
8.04	Problems involving bearings	U	F	PS		C
8.05	Pythagoras' theorem and trigonometry in 3D*	U	F	PS	R	C
8.06	The sine rule*	U	F	PS	R	
8.07	The sine rule for angles*	U	F	PS	R	C
8.08	The cosine rule*	U	F	PS	R	C
8.09	The cosine rule for angles*	U	F	PS	R	
8.10	The area of a triangle*	U	F	PS	R	
8.11	Problems involving the sine and cosine rules*	U	F	PS	R	C

*PATH

Wordbank

Quiz
Wordbank 8

angle of depression The angle of looking down, measured from the horizontal

angle of elevation The angle of looking up, measured from the horizontal

bearing The angle used to show the direction of one location from a given point

cosine rule A rule that relates the 3 sides and one of the angles of any triangle: $c^2 = a^2 + b^2 - 2ab\cos C$

included angle The angle between 2 known sides

minute (') A unit for measuring angle size, $\frac{1}{60}$ of a degree

sine rule A rule that relates the sides of any triangle to the sine of their opposite angles: $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$

theta (θ) A letter of the Greek alphabet used as a variable for angles

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (24):

SkillCheck Trigonometry • Trigonometry on a calculator • Rounding angles

8.01 The trigonometric ratios • Finding an unknown side • Finding the hypotenuse • Finding an unknown angle • Trigonometry

8.02 Trigonometry 2 • Angles of elevation and depression • Angle of depression 1 • Angle of depression 2

8.03 Three-figure bearings

8.04 Bearings • Trigonometry 2 • True bearings • Problems involving bearings

8.06 The sine rule

8.07 Obtuse angles using the sine rule • The sine rule

8.08 The cosine rule

8.09 The cosine rule for angles 2 • The cosine rule • The cosine rule for angles 1

8.10 The sine area formula

8.11 The sine and cosine rules

Quizzes (5):

- Wordbank 8
- SkillCheck 8
- Mental skills 8
- Language of maths 8
- Test yourself 8

Worksheets (15):

SkillCheck Trigonometric calculations • Investigating trigonometric ratios

8.01 Trigonometry review • Trigonometry problems

8.03 A page of bearings • NSW map bearings • 16 points of the compass

8.04 Elevations and bearings

8.06 Discovering the sine rule

8.07 Sine rule problems

8.09 Cosine rule problems

8.11 Finding an unknown side • Finding an unknown angle • The sine and cosine rules

Mind map Further trigonometry

Puzzles (10):

8.01 Trigonometry match-up • Trigonometric equations 1 • Trigonometry squaresaw • Trigonometry: Finding angles • Solving triangles • Trigonometry equations 2

8.03 Bearings match-up

8.05 Pythagorean two-step problems

8.11 The sine and cosine rules

Language of maths Further trigonometry crossword

Presentations (2):

8.02 Applications of triangles

8.05 2D and 3D applications of trigonometry

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- use trigonometry to solve problems involving right-angled triangles, to find unknown sides and angles in right-angled triangles where the angle is measured in degrees and minutes
- solve trigonometry problems involving angles of elevation and depression, and bearings
- (PATH) use Pythagoras' theorem and trigonometry to solve three-dimensional problems
- (PATH) use the sine and cosine rules to find an unknown side or angle in any triangle
- (PATH) use the formula $A = \frac{1}{2}ab\sin C$ to find the area of a triangle with side lengths a, b and included angle C

SkillCheck ANSWERS ON P. 611

1 Solve each equation.

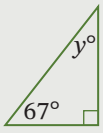
a $\frac{x}{5} = 7$ **b** $\frac{h}{4} = 8.3$ **c** $\frac{45}{y} = 9$

2 Find the value of each variable.

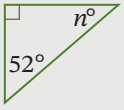
a



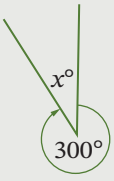
b



c



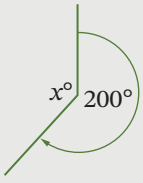
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e

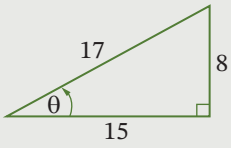


f

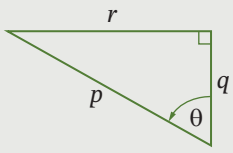


3 For each triangle, name the hypotenuse, opposite and adjacent sides for angle θ.

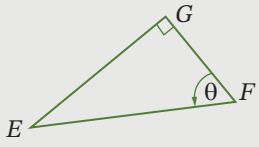
a



b



c



4 Evaluate each expression correct to 4 decimal places.

a cos 32° **b** sin 50.9° **c** 200 tan 18°

d tan 8° 45′ **e** 14 sin 87° 40′ **f** $\frac{13}{\cos 18° 27'}$

5 Round each angle to the nearest degree.

a 64° 27′ **b** 25° 43′ **c** 12° 8′ 50″

6 Round each angle to the nearest minute.

a 50° 19′ 26″ **b** 31° 55′ 55″ **c** 64° 18′ 30″

7 Convert each angle to degrees and minutes, correct to the nearest minute.

a 45.8° **b** 33.175° **c** 5.346°

Quiz
SkillCheck 8

Worksheets
Trigonometric calculations
Investigating trigonometric ratios

Videos
Trigonometry
Trigonometry on a calculator
Rounding angles

8

8.01 Right-angled trigonometry

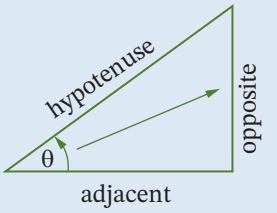
There are 3 special fractions called **trigonometric ratios** that relate the lengths of 2 sides of a right-angled triangle: **sine**, **cosine** and **tangent**.

Video The trigonometric ratios

Puzzle Trigonometry match-up

The trigonometric ratios

Ratio	Abbreviation	Meaning
sine	sin	$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}}$
cosine	cos	$\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}}$
tangent	tan	$\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$



You can remember the 3 ratios using **SOH-CAH-TOA** (pronounced 'so-car-toe-ah'):

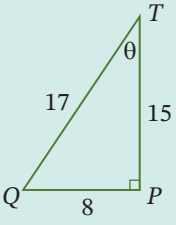
$$\sin = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{O}{H} \quad \text{SOH}$$

$$\cos = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{A}{H} \quad \text{CAH}$$

$$\tan = \frac{\text{opposite}}{\text{adjacent}} = \frac{O}{A} \quad \text{TOA}$$

Example 1

In $\triangle PTQ$, find $\sin\theta$, $\cos\theta$ and $\tan\theta$.



Solution

For angle θ, opposite = 8, adjacent = 15, hypotenuse = 17.

$$\sin\theta = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{8}{17} \qquad \cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{15}{17} \qquad \tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{8}{15}$$

Videos Finding an unknown side; Finding the hypotenuse; Trigonometry

Finding an unknown side

We can use a trigonometric ratio to calculate the length of an unknown side in a right-angled triangle if one other side and angle are known. We need to select the correct ratio that links the given angle to the unknown side and known side.

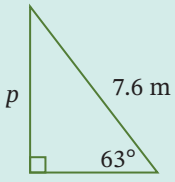
Finding an unknown side in a right-angled triangle

1. Identify the 2 labelled sides and decide whether to use sin, cos or tan
2. Write an equation using the ratio, the given angle and the variable
3. Solve the equation to find the value of the variable

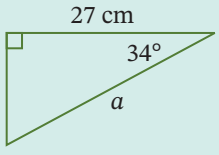
Example 2

Find the value of each variable, correct to one decimal place.

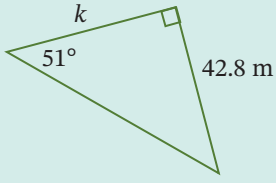
a



b



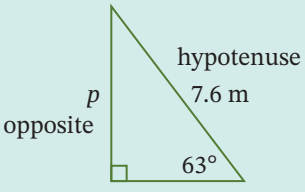
c



Solution

a SOH, CAH or TOA?

The marked sides are the opposite (O) side and the hypotenuse (H), so use sin.



$$\sin 63° = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$= \frac{p}{7.6}$$

$$\sin 63° \times 7.6 = \frac{p}{7.6} \times 7.6$$

Multiplying both sides by 7.6

$$7.6 \sin 63° = p$$

$$p = 7.6 \sin 63°$$

Swapping sides

$$= 6.7716...$$

$$\approx 6.8$$

From the diagram, a length of 6.8 m looks reasonable.

b 27 cm is adjacent, a is hypotenuse, so use cos.

$$\cos 34° = \frac{27}{a}$$

Note that the variable a appears in the denominator of the equation

$$\cos 34° \times a = \frac{27}{a} \times a$$

Multiplying both sides by a

$$a \cos 34° = 27$$

$$\frac{a \cos 34°}{\cos 34°} = \frac{27}{\cos 34°}$$

Dividing both sides by cos 34°

$$a = \frac{27}{\cos 34°}$$

$$= 32.5678...$$

$$\approx 32.6 \text{ cm}$$

Note that when the unknown appears in the denominator of an equation, it can swap positions with the trigonometric ratio, so that $\cos 34° = \frac{a}{27}$ becomes $a = \frac{27}{\cos 34°}$.

c 42.8 m is opposite, k is adjacent, so use tan.

$\tan 51° = \frac{42.8}{k}$ k appears in the denominator

$k = \frac{42.8}{\tan 51°}$ Swapping the position of k with tan 51°

$= 34.6587...$

≈ 34.7 m

Alternative method

To avoid having k in the denominator, we could use tan with the third angle of the triangle.

3rd angle $= 180° - 90° - 51°$

$= 39°$

$\tan 39° = \frac{k}{42.8}$

$k = 42.8 \tan 39°$

$= 34.6587...$

≈ 34.7 m

Videos
Finding an unknown angle
Trigonometry

Finding an unknown angle

A scientific calculator can also be used to find an unknown angle, θ, if the trigonometric ratio of the angle is known, for example, if $\sin \theta = 0.9063$.

An unknown angle can be found using the $\sin^{-1}$, $\cos^{-1}$ and $\tan^{-1}$ keys on the calculator. These are the **inverse sin**, **inverse cos** and **inverse tan** functions, found by pressing the SHIFT or 2nd F key before the sin, cos or tan key.

Worksheets
Trigonometry review
Trigonometry problems

Puzzles
Trigonometry squaresaw
Trigonometry: Finding angles
Solving triangles
Trigonometry equations 2

Example 3

a If $\sin \alpha = \frac{5}{8}$, find the value of angle α, correct to the nearest degree.

b If $\tan X = 2.1572$, find the value of angle X, correct to the nearest minute.

Solution

a $\sin \alpha = \frac{5}{8}$

$\alpha = 38.6821...°$ On a calculator: SHIFT sin 5 ▭ 8 =

$\approx 39°$ Note: ▭ or $a^{b/c}$ is the fraction key

b $\tan X = 2.1572$ SHIFT tan 2.1572 =

$X = 65.1292...$

$= 65° 7' 45.35''$ °’” or DMS

$\approx 65° 8'$

Finding an unknown angle in a right-angled triangle

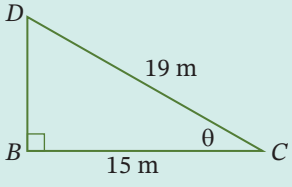
1. Identify 2 known sides and decide whether to use the sin, cos or tan ratio.
2. Write an equation using the ratio, the angle and the 2 sides as a fraction.
3. Use the calculator's inverse trigonometric function to find the size of the angle.



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Example 4

Find the size of angle θ, correct to the nearest degree.

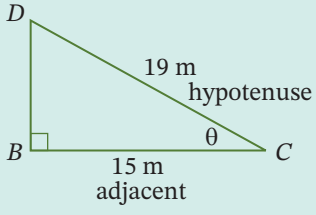


Solution

SOH, CAH or TOA?

The known sides are the adjacent (A) side and the hypotenuse (H), so use cos.

$\cos\theta = \frac{15}{19}$



$\theta = 37.8636...$

$\approx 38°$

On a calculator: SHIFT cos 15 ▭ =

From the diagram, an angle size of 38° looks reasonable.

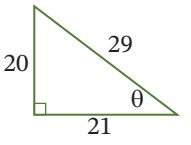
EXERCISE 8.01 ANSWERS ON P. 611

Right-angled trigonometry U F PS C

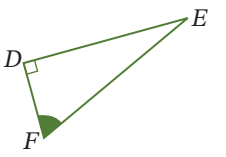
EXAMPLE 1

1 For each marked angle, find the sine, cosine and tangent ratios. C

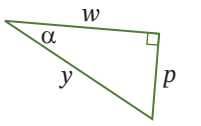
a



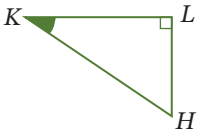
b



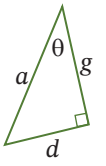
c



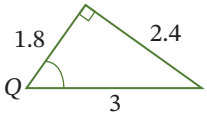
d



e



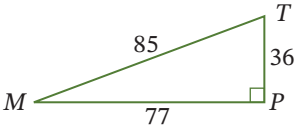
f



2 For the triangle shown, find:

a $\cos T$ **b** $\tan M$

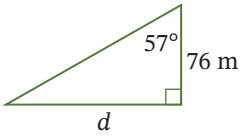
c $\sin T$ **d** $\sin M$



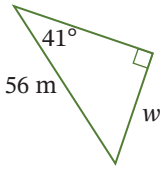
EXAMPLE 2

3 Find the value of the variable in each triangle, correct to one decimal place.

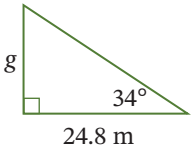
a



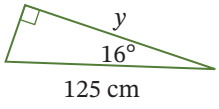
b



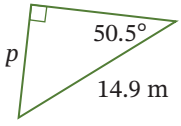
c



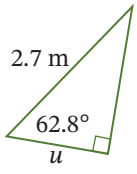
d



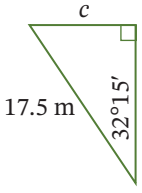
e



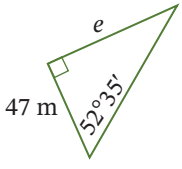
f



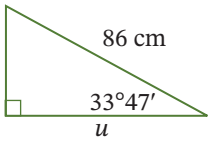
g



h



i



EXAMPLE 3

4 Find the size of angle θ, correct to the nearest degree.

a $\tan\theta = 1.85$ **b** $\sin\theta = 0.6432$ **c** $\cos\theta = \frac{\sqrt{3}}{2}$ **d** $\tan\theta = 7.1$

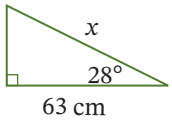
e $\sin\theta = \frac{3}{4}$ **f** $\cos\theta = \frac{2}{17}$ **g** $\tan\theta = \frac{1}{\sqrt{3}}$ **h** $\sin\theta = \frac{1}{\sqrt{2}}$

Foundation Standard Complex

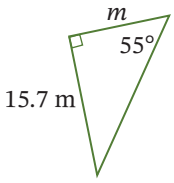
9780170479622

5 Find the value of each variable, correct to one decimal place.

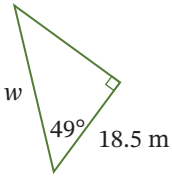
a



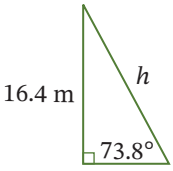
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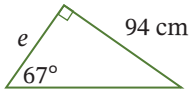
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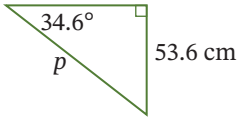
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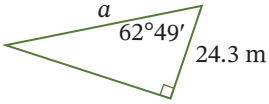
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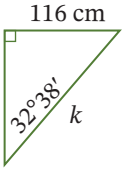
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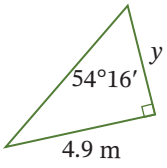
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h

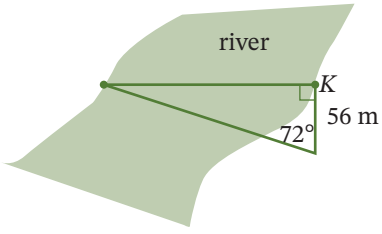


i

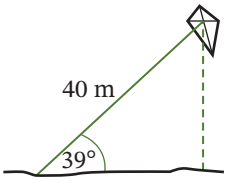


6 Find each length or distance, correct to one decimal place. PS

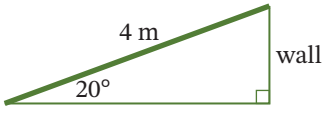
a The width of the river at the point K



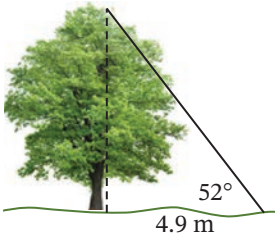
b The height of the kite



c The distance the base of the ramp is from the wall.

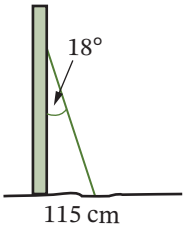


d The height of the tree

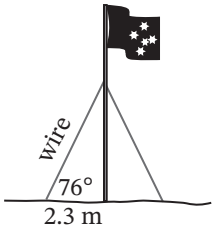


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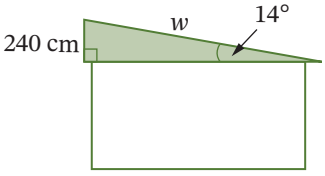
e The distance the ladder reaches up the wall.



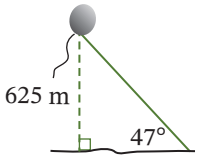
f The length of the support wire.



g The length of the sloping roof.



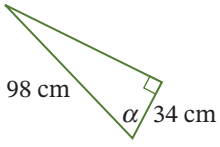
h The distance a balloon has drifted from its starting point.



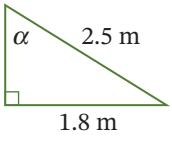
EXAMPLE 4

7 Find the size of angle α, correct to the nearest minute.

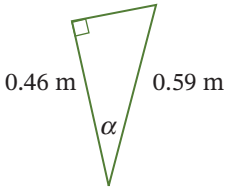
a



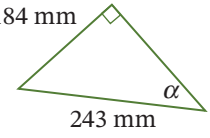
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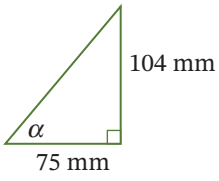
c



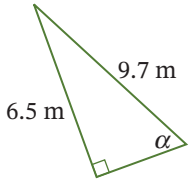
d



e

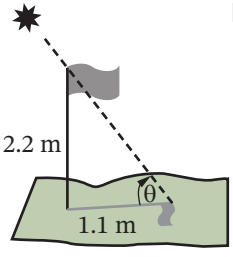


f

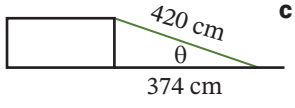


8 Find the size of angle θ, correct to the nearest degree.

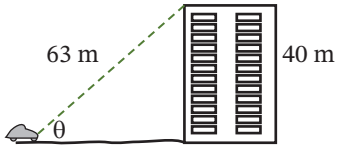
a



b



c



9 A ladder 4.3 m long is leaning against a wall. If the angle between the ladder and the wall is 28°, how far up the wall does the ladder reach? PS

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10 The entrance to a bank has a ramp for wheelchair access that is 3.6 m long. If the ramp is inclined at an angle of 10° to the ground, what is the height of the entrance (to the nearest cm)? PS

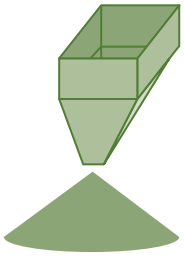
For questions **11** to **15**, give your answers correct to the nearest degree.

11 A ladder 6.8 m long is placed against a wall. If the foot of the ladder is 2.1 m from the base of the wall, find the angle at which the ladder is inclined to the ground. PS

12 A walking trail falls 7 m for every 130 m travelled along the trail. At what angle is the trail inclined to the horizontal? PS

13 A tree that is 15 m tall casts a shadow that is 19.6 m long. Find the angle of the sun from the ground. PS

14 When sand is poured from a hopper at a steady rate it forms a conical pile. The height of the pile is 3 m and its diameter is 8.6 m. Calculate the angle of repose of the sand (the angle the sloping side makes with the horizontal base) PS



15 Calculate the take-off angle of a passenger plane if its altitude after 30 seconds is 600 m and it has flown a distance of 2500 m. PS



Shutterstock.com/ Ryan Fletcher

16 A rectangular gate has a diagonal brace that makes an angle of 35° with the top of the gate. The height of the gate is 1200 mm. Calculate the length of the diagonal brace. Select the correct answer **A**, **B**, **C** or **D**. PS

A 1465 mm **B** 1714 mm **C** 2092 mm **D** 2134 mm

17 A stretch of road rises 220 m for every 3 km travelled along the road. Find the angle at which the road is inclined to the horizontal. PS

18 A roof has a pitch of 18° 26′. The height of the roof is 4.1 m. Calculate the overall width of the roof (correct to one decimal place). PS



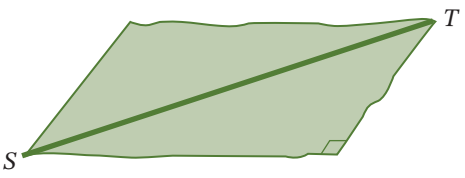
19 A glider is descending at an angle of 25° to the horizontal. The length of its flight path until it lands is 3.5 km. What was the altitude (to the nearest 0.1 km) of the glider? PS



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20 The diagonal path of a large rectangular reserve makes an angle of 32° with the longer side of the reserve. The width of the reserve is 230 m. PS



a Calculate the length of the path (correct to the nearest metre).

b How much shorter is walking along the path from S to T than walking along the 2 edges of the reserve?

21 $\triangle PQR$ is right-angled at Q, $PR = 15.3$ m and $\angle P = 23°$. Find the length of QR, correct to 2 decimal places.

22 $\triangle TPM$ is right-angled at M, $MT = 39.3$ cm and $\angle T = 44.7°$. Find the length of PT, correct to one decimal place.

23 In $\triangle WYX$, $\angle Y = 90°$, $\angle W = 74°\ 25'$ and $XY = 245$ mm. Find the length of WY, correct to the nearest millimetre.

24 $\triangle KLM$ is right-angled at L, $KL = 19$ cm and $LM = 21$ cm. Find $\angle K$, correct to the nearest degree.

25 In $\triangle PTW$, $\angle T = 90°$, $PW = 22.3$ m and $TW = 7.6$ m. Find $\angle W$, correct to the nearest degree.

26 In $\triangle ABC$, $\angle B = 90°$, $BC = 0.8$ m and $AC = 1.3$ m. Find $\angle C$, correct to the nearest minute.

Did you know?

The Greek alpha-bet

Here are 8 letters (in lower-case and capitals) from the Greek alphabet.

α, A alpha	β, B beta	γ, Γ gamma	δ, Δ delta
θ, Θ theta	φ, Φ phi	σ, Σ sigma	ω, Ω omega

The ancient Greeks had a great influence on the development of mathematics. It is traditional to use Greek letters as variables, particularly in geometry and trigonometry.

1 **Find out how many letters there are in the Greek alphabet, and name each one.**

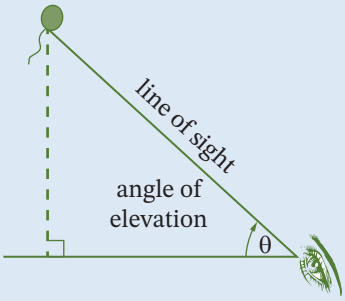
2 **Compare the Greek alphabet with our Roman alphabet.**

3 **Can you see where the word *alphabet* comes from? Explain how it originated.**

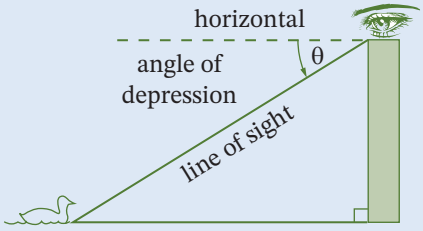
Angles of elevation and depression 8.02

Angles of elevation and depression

The **angle of elevation** is the angle of looking **up**, measured from the horizontal.



The **angle of depression** is the angle of looking **down**, measured from the horizontal.



Problems involving angles of elevation and depression usually require the tan ratio in their solutions.

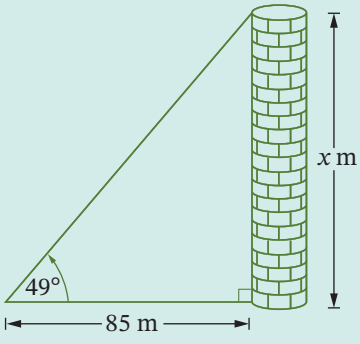
An instrument for measuring an angle of elevation or depression is a **clinometer**. It is like a protractor with a sighting tube attached.



Getty Images/VW Pics

Example 5

The angle of elevation of a chimney is 49° at a distance of 85 m from its base. Find the height of the chimney, to the nearest metre.



Solution

Let the height be x metres.

$$\tan 49° = \frac{x}{85}$$

$$x = 85 \tan 49°$$

$$= 97.7813...$$

$$\approx 98$$

The height of the chimney is 98 m.

Videos
Trigonometry 2
Angles of elevation and depression

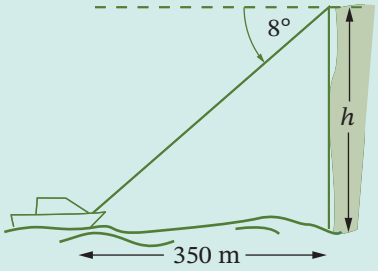
Presentation
Applications of triangles

8.02

Videos
Angles of elevation and depression
Angle of depression 1
Angle of depression 2

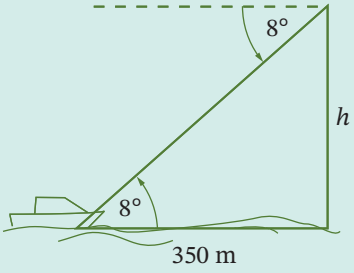
Example 6

The angle of depression of a boat from the top of a cliff is 8°. If the boat is 350 m from the base of the cliff, calculate the height of the cliff, correct to the nearest metre.



Solution

By alternate angles on parallel lines, the angle of elevation of the top of the cliff from the boat is also 8°.



$$\tan 8° = \frac{h}{350}$$
$$h = 350 \tan 8°$$
$$= 49.1892...$$
$$\approx 49$$

The height of the cliff is 49 m.

Alternative method:

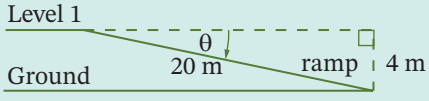
The third angle in the triangle (adjacent to the angle of depression) = 90° – 8° = 82°.

$$\tan 82° = \frac{350}{h}$$
$$h = \frac{350}{\tan 82°}$$
$$= 49.1892...$$
$$\approx 49$$

The height of the cliff is 49 m.

Example 7

The ramp from one level to the next in a car park is 20 m long and drops 4 m. Find the angle of depression of the ramp, to the nearest degree.



Solution

$$\sin\theta = \frac{4}{20}$$
$$\theta = 11.5369...°$$
$$\approx 12°$$

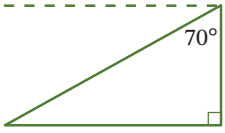
The angle of depression of the ramp is 12°.

EXERCISE 8.02 ANSWERS ON P. 612

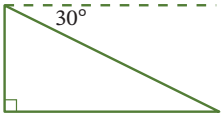
Angles of elevation and depression U F PS

1 Copy each diagram, mark the angle of elevation θ and find its size.

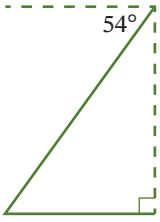
a



b

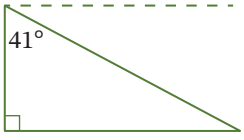


c

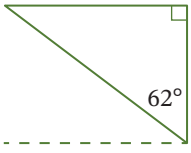


2 Copy each diagram, mark the angle of depression θ and find its size.

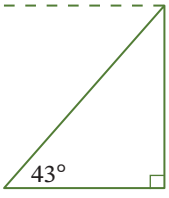
a



b

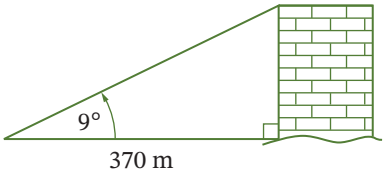


c

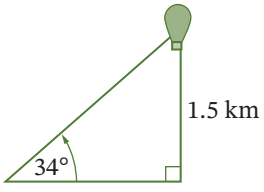


3 Millie stands 370 m from the base of a building. Using a clinometer, she finds that the angle of elevation of the top is 9°. Find the height of the building, to the nearest metre.

EXAMPLE 5

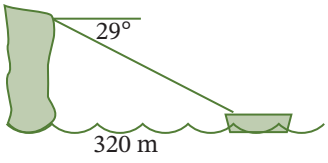


4 The angle of elevation of a weather balloon at a height of 1.5 km is 34°. How far (to the nearest metre) is the observer from being directly under the balloon?

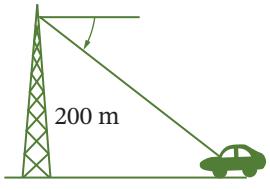


5 A raft is 320 m from the base of a cliff. The angle of depression of the raft from the top of the cliff is 29°. Find the height of the cliff, to the nearest metre. PS

EXAMPLE 6



6 From the top of a 200 m tower, the angle of depression of a car is 48°. How far is the car from the foot of the tower? PS



7 A 275 m radio mast is 1.7 km from a school. Find, correct to the nearest degree, the angle of elevation of the top of the mast from the school. PS

EXAMPLE 7

8 A monument 24 m high casts a shadow 20 m long. Calculate, correct to the nearest degree, the angle of elevation of the sun at this time of day. PS

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9 In a concert hall, Ethan is sitting 20 m from the stage by line of sight. He is also 5 m above the level of the stage. At what angle of depression is the stage? Answer to the nearest minute. PS

10 A plane is 340 m directly above one end of a 1000 m runway. Find, correct to the nearest minute, the angle of depression to the far end of the runway. PS



verzellenberg/ Shutterstock.com

11 An observer 174 cm tall is standing 11.6 m from the base of a flagpole. The angle of elevation to the top of the flagpole is 43°. How high is the flagpole, to the nearest cm? PS

12 A flagpole is mounted on top of a tall building. At a distance of 250 m from the base of the building, the angles of elevation of the bottom and top of the flagpole are 38° and 40° respectively. Calculate the height of the flagpole, correct to one decimal place. PS

13 A news helicopter hovers at a height of 500 m. The angles of depression of a fire moving in the direction of the helicopter are first 10° and then 15°. How far (to the nearest metre) has the fire moved between the 2 observations? PS

14 The angle of elevation to the bottom of a transmission tower on a hill from an observer 1.8 km away from the base of the hill is 5°. The angle of elevation to the top of the tower from the observer is 6.8°. Find the height of the tower to the nearest metre. PS

8.03 Bearings

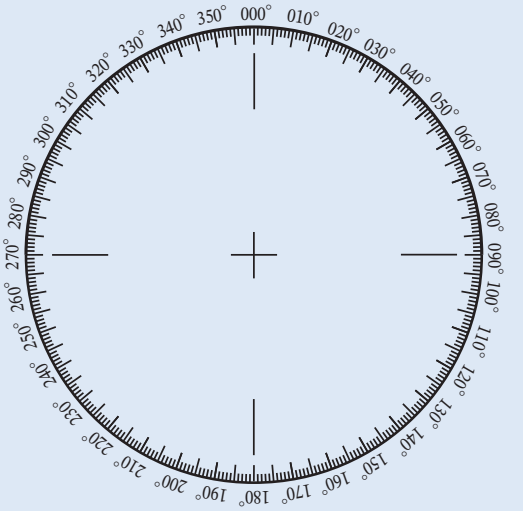
Bearings are used in navigation. A **bearing** is an angle measurement used to describe precisely the direction of one location from a given reference point.

Worksheets
A page of bearings
NSW map bearings
16 points of the compass

Puzzle
Bearings match-up

True bearings

True bearings, also called **three-figure bearings**, use angles from 000° to 360° to show the amount of turning measured **clockwise from north 000°**. Note that the angles are always written with 3 digits.



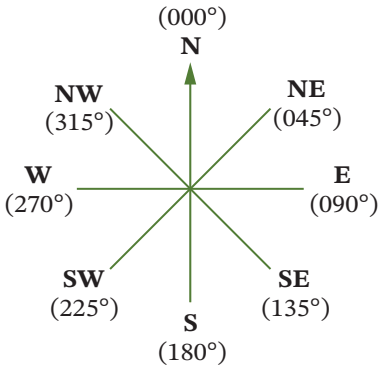
True bearings

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This **compass rose** shows the true bearings of 8 points on the compass. A bearing of due east is 090°, while a compass direction of southwest (SW) is 225°.

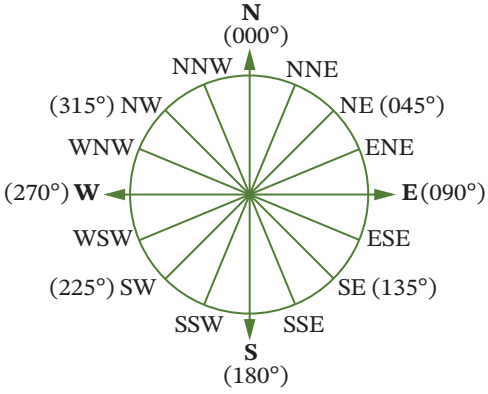
- Bearings from 000º to 090º are in the NE quadrant
- Bearings from 090º to 180º are in the SE quadrant
- Bearings from 180º to 270º are in the SW quadrant
- Bearings from 270º to 360º are in the NW quadrant



8.03

This diagram shows the 16 points of the compass.

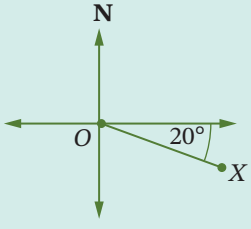
N = north	NE = northeast
E = east	SE = southeast
S = south	SW = southwest
W = west	NW = northwest
NNE = north-northeast	ENE = east-northeast
ESE = east-southeast	SSE = south-southeast
SSW = south-southwest	WSW = west-southwest
WNW = west-northwest	NNW = north-northwest



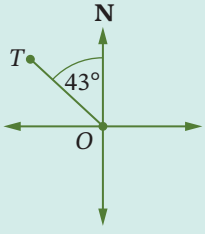
Example 8

Find the true bearing of each point from O.

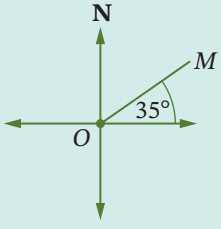
a



b



c



Video
Three-figure bearings

Solution

a Bearing of X from O is $90° + 20° = 110°$.

b Bearing of T from O is $360° - 43° = 317°$.

c Bearing of M from O is $90° - 35° = 055°$

This is sometimes written as 110°T to indicate that it is a True bearing.

Must be written as a 3-digit angle

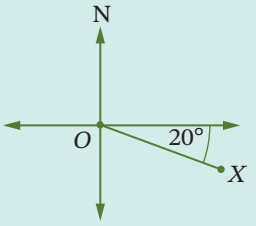
Compass bearings

Compass bearings use acute angles measured from the North-South axis towards East or West, such as N20°E or S57°W. They are named according to the quadrant in which they are located, and are used in land surveying.

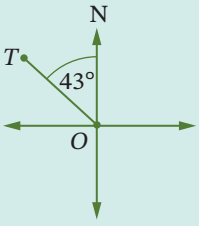
Example 9

Write the compass bearing of each point from O.

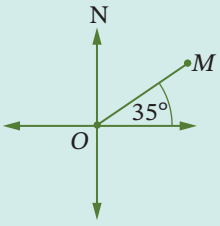
a



b



c



Solution

a X is in the southeast (SE) quadrant. From south, it is $90° - 20° = 70°$ towards east.

The compass bearing of X from O is S70°E.

b T is in the northwest (NW) quadrant. From north, it is 43° towards west.

The compass bearing of T from O is N43°W.

c M is in the northeast (NE) quadrant. From north, it is $90° - 35° = 55°$ towards east.

The compass bearing of M from O is N55°E.

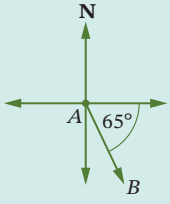
Video
Three-figure bearings

Example 10

Sketch point B on a compass rose if B has a bearing of 155° from A.

Solution

Draw the compass rose on the point where the bearing is measured from.
155° is between 90° and 180°, so B is in the southeast (SE) quadrant.
$155° - 90° = 65°$, so B is 65° from east (E).



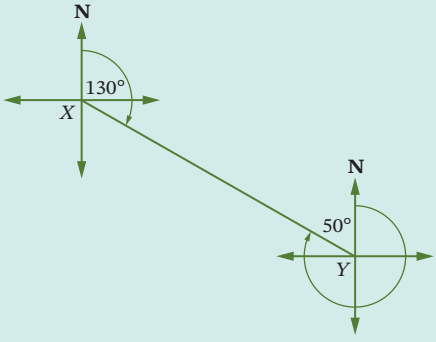
Video
Three-figure bearings

Example 11

The bearing of Y from X is 130°. What is the bearing of X from Y?

Solution

Sketch the bearing of Y from X.
On the same diagram, draw a compass rose at Y and find $\angle NYX$.
$\angle NYX = 50°$ (co-interior angles, $NX \parallel NY$)
$\therefore$ Bearing of X from $Y = 360° - 50°$
$= 310°$



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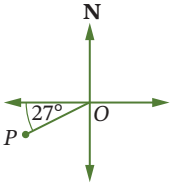
EXERCISE 8.03 ANSWERS ON P. 612

Bearings UFRC

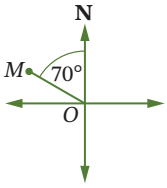
EXAMPLE 8

1 Find the true bearing of each point from O. C

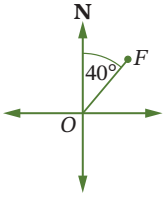
a



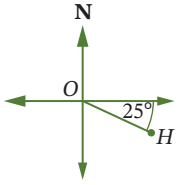
b



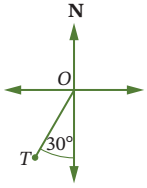
c



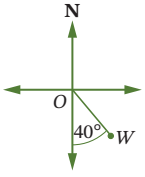
d



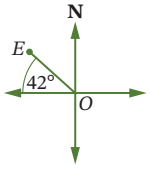
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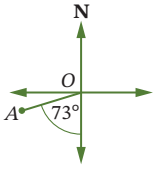
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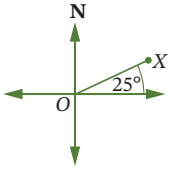
g



h

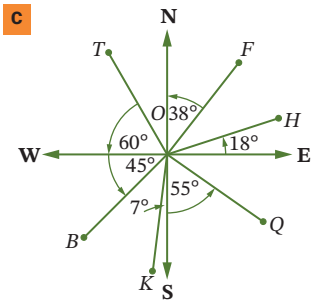


i



2 What is the true bearing of each direction or point from O? C

a N	**b** E	**c** S
d W	**e** F	**f** Q
g T	**h** B	
i H	**j** K	



EXAMPLE 9

3 Find the compass bearing of each point from O in Question **1**. C

4 Find the true bearing of each compass direction. C

a SW	**b** E	**c** NE	**d** ESE
e NNW	**f** W	**g** SE	**h** WSW

5 Sketch each bearing on a compass rose.

a 220°	**b** 060°	**c** 260°	**d** 125°
e N10°W	**f** S87°W	**g** S9°E	**h** N32°E

6 **a** What is the compass direction halfway between northwest and west?

b What is the compass bearing of this compass direction? C

EXAMPLE 10

7 Sketch P on a compass rose if P has a bearing of:

a 132° from T	**b** 260° from M	**c** 335° from X	**d** 010° from K

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EXAMPLE 11

8 If the bearing of P from A is 060°, what is the bearing of A from P? R

9 The bearing of T from Y is 100°. What is the bearing of Y from T? R

10 What is the (smallest) angle between:

a S and SW? **b** NE and SE? **c** E and NW?

d NE and SSW? **e** E and SSW? **f** SW and WNW?

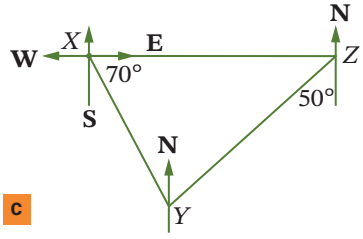
11 The compass bearing of H from M is N65°W. Find the compass bearing of M from H. R C

12 Draw a diagram for each situation described. C

a A plane flies on a bearing of 280° for 150 km and then another 250 km on a bearing of 080°.

b A cyclist travels 15 km due east and then 20 km on a S45°W bearing.

13 For this diagram, find the true bearing of: C



a Y from Z

b X from Z

c Y from X

14 For the above diagram, find the compass bearing of: C

a X from Y

b Z from Y

c Z from X

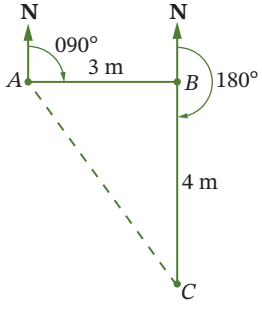
Investigation

Compass walks

You need: a directional compass and a tape measure or trundle wheel.

This activity can also be done in the classroom using scale drawings, graph paper, a ruler and a protractor.

A triangular walk



1 Starting at A, walk due east for 3 m to B.

2 From B, walk due south for 4 m to C.

3 How far is C from A?

4 What is the bearing of:

a A from C?

b C from A?

A square walk

1 Starting at P, walk at a bearing of 045° for 8 m to Q.

2 From Q, walk at a bearing of 315° for 8 m to R.

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3 From R, walk at a bearing of 225° for 8 m to S.

4 How far is S from P?

5 What is the bearing of:

a P from S? b S from P?

A pentagonal walk

1 Starting at U, walk at a bearing of S50°E for 4 m to V.

2 From V, walk at a bearing of N40°E for 7 m to W.

3 From W, walk at a bearing of N40°W for 4.8 m to X.

4 From X, walk at a bearing of west for 4.5 m to Y.

5 How far is Y from U?

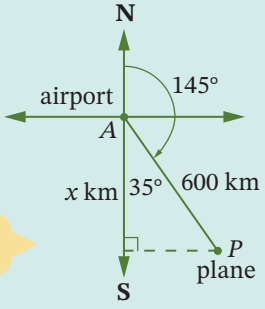
6 What is the bearing of:

a U from Y? b Y from U?

Problems involving bearings 8.04

Example 12

A plane leaves an airport and flies 600 km on a bearing of 145°. How far south of the airport is the plane (to the nearest km)?



Solution

To find x, create a right-angled triangle.

Let x km = distance south

$\therefore \angle SAP = 180° - 145°$ (angles on a straight line)

$= 35°$

$\cos 35° = \frac{x}{600}$

$x = 600 \cos 35°$

$= 491.4912...$

≈ 491

The plane is 491 km south of the airport.

Worksheet
Elevations and bearings

Videos
Bearings
Trigonometry 2
True bearings

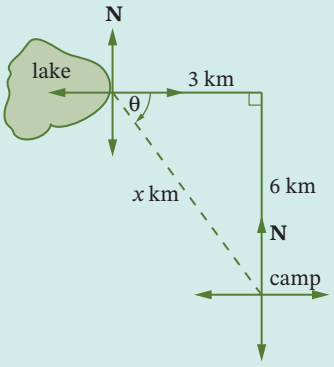
Video
Problems involving bearings

Example 13

From camp, Sumaiyaa walks due north for 6 km, then 3 km due west to a lake.

a How far is Sumaiyaa from the camp, correct to the nearest 0.1 km?

b What is the bearing of the camp from the lake (to the nearest minute)?



Solution

a Let x = distance from camp.

$x^2 = 6^2 + 3^2$ by Pythagoras' theorem

$= 45$

$x = \sqrt{45}$

$= 6.708...$

≈ 6.7 km

Sumaiyaa is 6.7 km from the camp.

b Note angle θ in the diagram.

$\tan\theta = \frac{6}{3}$

$\theta = 63.434...°$

$= 63° 26' 5.82"$

$\approx 63° 26'$

Bearing of camp from lake

$= 90° + 63° 26'$

$= 153° 26'$

EXERCISE 8.04 ANSWERS ON P. 613

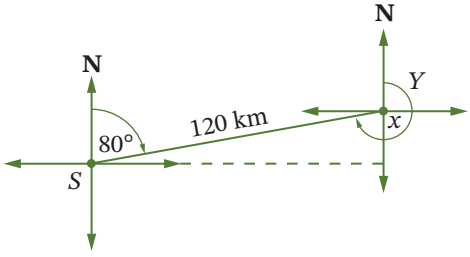
Problems involving bearings U F PS C

EXAMPLE 12

1 A yacht leaves Sydney and sails 120 km on a bearing of 080°.

a How far north (to the nearest km) of Sydney is the yacht?

b What is the bearing of Sydney from the yacht?

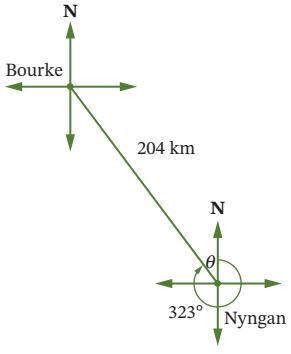


2 Declan leaves Nyngan and drives 204 km to Bourke. The bearing of Bourke from Nyngan is 323°.

a Find the value of θ.

b How far north (to the nearest km) of Nyngan is Bourke?

c What is the bearing of Nyngan from Bourke?



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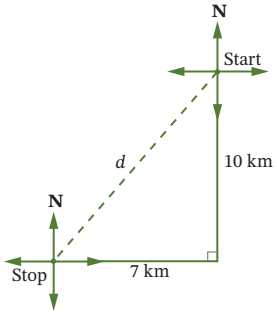
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3 The distance 'as the crow flies' from Sydney to Wollongong is 69 km. If the bearing of Wollongong from Sydney is 205°, calculate:

- **a** how far south Wollongong is from Sydney (correct to one decimal place)
- **b** how far east Sydney is from Wollongong (correct to one decimal place)
- **c** the compass bearing of Sydney from Wollongong.

4 Jana cycles 10 km due south, then 7 km due west.

- **a** How far (correct to one decimal place) is Jana from her starting point?
- **b** What is her true bearing from the starting point, correct to the nearest degree?



5 A triathlete cycles 20 km on a SSE bearing to the finish line.

- **a** How far (to the nearest km) has the triathlete travelled in a southerly direction?
- **b** What is the compass bearing of the starting point from the finish line? PS C

6 A hiking group walks from Sandy Flats to Black Ridge (a distance of 20.9 km) in the direction 078°. They then turn and hike due south to Rivers End, then due west back to Sandy Flats. How far have they hiked altogether (to the nearest 0.1 km)? PS C

7 A triangular orienteering run starts at Alpha and passes through the checkpoints of Bravo and Charlie before finishing at Alpha. Bravo is 8.5 km due east of Alpha, and Charlie is 10.5 km due south of Bravo. PS C

- **a** Calculate, correct to 3 decimal places, the distance from Charlie to Alpha.
- **b** Find the compass bearing of Alpha from Charlie to the nearest degree.

8 A plane takes off from Darwin at 10:15 a.m. and flies on a bearing of 150° at 700 km/h. PS C

- **a** How far (to the nearest km) due south of Darwin is the plane at 1:45 p.m.?
- **b** What is the bearing (correct to the nearest degree) of Darwin from the plane?

9 A fishing trawler sails 30 km from port on a bearing of S60°E until it reaches a submerged reef. How far (to the nearest km) is the port: PS C

- **a** north of the reef?
- **b** west of the reef?

10 2 racing pigeons are set free at the same time. The first bird flies on a course of N40°E while the second bird flies on a course of S50°E. PS C

- **a** The first bird flies 200 km until it is due north of the second bird. Find their distance apart, correct to 2 decimal places.
- **b** How far has the second bird flown?

11 2 horse riders start from the same stable. The rider of the black horse goes due west for 5.5 km and stops. The rider of the chestnut horse travels in a direction of N57°W until he is due north of the black horse. How far did the rider of the chestnut horse travel? Answer correct to 2 significant figures. PS C

12 2 ships leave from the same port. One ship travels on a bearing of 157° at 20 knots. The second ship travels on a bearing of 247° at 35 knots. (1 knot is a speed of 1 nautical mile per hour.) PS C

a How far apart are the ships after 8 hours, to the nearest nautical mile?

b Calculate the bearing of the second ship from the first, to the nearest minute.

13 A plane flies 1200 km on a NE compass bearing and then changes direction to a bearing of NW and flies for a distance of 450 km. PS C

a How far, correct to the nearest km, is the plane from its starting point?

b What is the compass bearing of the plane from its starting point, to the nearest minute?

14 A fishing boat leaves Sydney Harbour on a bearing of ESE and after 15 nautical miles, changes direction to a bearing of SSW and travels a further 10 nautical miles.

Find the true bearing of Sydney from the boat's final position, correct to the nearest minute. PS C

8.05 Pythagoras' theorem and trigonometry in 3D

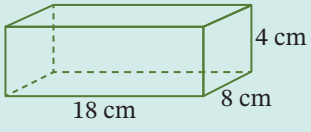
PATH

Puzzle
Pythagorean two-step problems

Presentation
2D and 3D applications of trigonometry

Example 14

A wooden rectangular box has dimensions 18 cm × 8 cm × 4 cm.



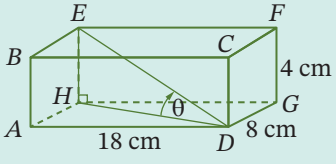
a Find, correct to one decimal place, the length of the longest pencil that can lie flat in the base of the box.

b Find, correct to one decimal place, the length of the longest diagonal of the box.

c Find, correct to the nearest degree, the angle that the longest diagonal makes with the base of the box.

Solution

Label the box as shown. *HD* is the length of the longest pencil that can lie flat in the base of the box, while *ED* is the longest diagonal of the box.



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a Using the right-angled triangle DAH:

$HD^2 = DA^2 + AH^2$
$= 18^2 + 8^2$
$= 388$
$HD = \sqrt{388}$
$= 19.6977...$
≈ 19.7 cm

The longest pencil that can lie flat in the base of the box is 19.7 cm.

b $ED^2 = HD^2 + HE^2$
$ED^2 = \left(\sqrt{388}\right)^2 + 4^2$
$= 404$
$ED = \sqrt{404}$
$= 20.0997...$
≈ 20.1 cm

The longest diagonal of the box is 20.1 cm.

c In the right-angled triangle EHD, θ is the angle that the longest diagonal makes with the base of the box.

$\tan\theta = \frac{EH}{HD}$
$= \frac{4}{\sqrt{388}}$
$\theta = 11.4789...°$
$\approx 11°$

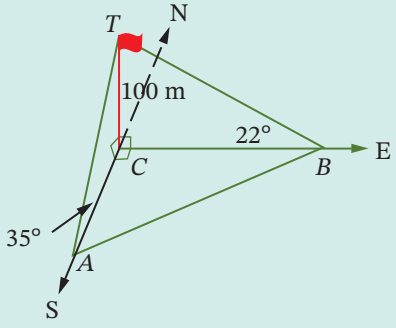
The longest diagonal makes an angle of 11° with the base of the box.

or use sin or cos since ED is also known

4 ▦ √ 388 SHIFT tan =

Example 15

A 100 m flagpole is observed from 2 different locations. From point A, due south of the flagpole the angle of elevation of the top of the flagpole is 35°; from point B, due east of the flagpole the angle of elevation is 22°. Find the distance between A and B, correct to the nearest metre.



Solution

There are 3 right-angled triangles in this diagram. To find AB, we must first find AC and CB using trigonometry.

In $\triangle ACT$,

$\tan 35° = \frac{100}{AC}$
$AC = \frac{100}{\tan 35°}$
$= 142.8148...$

Do not round yet.

In $\triangle BCT$,

$\tan 22° = \frac{100}{CB}$
$CB = \frac{100}{\tan 22°}$
$\approx 247.5086...$

In $\triangle ABC$,

$AB^2 = AC^2 + CB^2$
$= (142.8148...)^2 + (247.5086...)^2$
$= 81656.6166...$
$AB = \sqrt{81656.6166...}$
$= 285.7562...$
≈ 286 m

From the diagram, 286 m seems to be a reasonable answer.

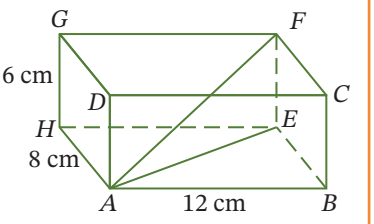
PATH

EXERCISE 8.05 ANSWERS ON P. 613

Pythagoras' theorem and trigonometry in 3D U F PS R C

EXAMPLE 14

1 For this rectangular prism, calculate:

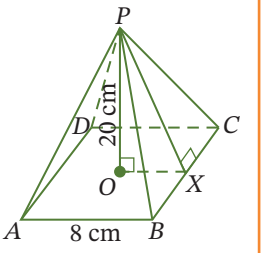


- **a** the length of AE in surd form
- **b** the length of AF, correct to one decimal place
- **c** the size of $\angle FAE$, correct to the nearest degree.

2 Sketch a diagram of a cube of length 20 cm, then find:

- **a** the exact length of the longest diagonal on any face
- **b** the length of the longest diagonal of the cube, correct to 2 decimal places
- **c** the angle that the longest diagonal makes with the base, correct to the nearest minute.

3 The diagram shows a square pyramid with base length 8 cm and perpendicular height 20 cm. PX is the slant height of the pyramid. Calculate: R



- **a** the length of PX, correct to 2 decimal places
- **b** the angle of inclination of PX, correct to the nearest degree.

4 A cone has a base diameter of 2.8 m and a slant height of 2.5 m. Find the angle that the cone makes with the vertical at the top of the cone. R

5 A fruit juice container is a square prism with dimensions 8 cm by 3 cm by 3 cm. R

- **a** Find, correct to one decimal place, the length of the longest drinking straw that fits inside the container.
- **b** To the nearest degree, what angle does the longest straw make with the vertical?

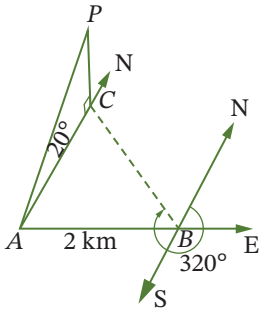
6 From a point X, 37 m from the base of a tree, the angle of elevation is 55° while the angle of elevation of the tree, from a point Y due east of the tree, is 25°. Find, correct to the nearest metre: PS R

- **a** the height of the tree
- **b** the distance of the tree from point Y

EXAMPLE 15

7 A 15 m flagpole stands on level ground. From point P, due west of the flagpole the angle of elevation of the top of the pole is 38°. From point Q, due north of the flagpole, the flagpole has an angle of elevation of 25°. Find PQ, correct to one decimal place. PS R

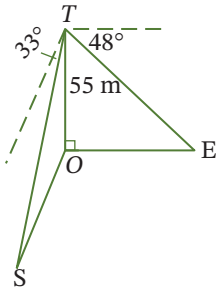
8 From a point, A, at the base of a mountain, the mountain peak, P, is due north and has an angle of elevation of 20°. From a point, B, 2 km due east of A on the same level, the mountain peak has a bearing of 320°. PS R



a What is the size of $\angle CBA$?

b Calculate the height, PC, of the mountain correct to the nearest metre.

9 From the top of her 55-metre office building, Madison observes 2 cars parked at ground level. The angle of depression of the car due east of the building is 48° and the angle of depression of the car parked due south of the building is 33°. PS R



Calculate, correct to the nearest metre, how far:

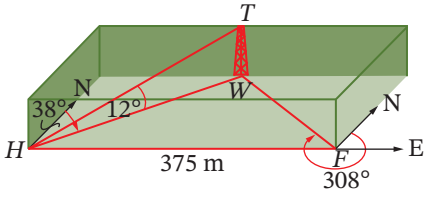
a each car is from the building

b the cars are apart.

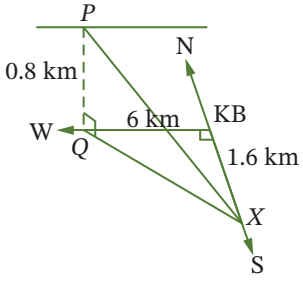
10 Hassan observes a transmission tower at an elevation angle of 12° and bearing 038°. Fatima stands 375 m due east of Hassan and observes the tower at a bearing of 308°. PS R

a Find the sizes of the angles of $\triangle FHW$.

b Find the height of the tower TW, correct to one decimal place.



11 A plane flies 6 km due west of Keira Bay at a constant height of 800 m. Xander sees the plane from his house 1.6 km south of Keira Bay.



Find, correct to the nearest degree: PS R C

a the bearing of the plane from Xander

b the angle of elevation of the plane from Xander.

Presentation
2D and 3D applications of trigonometry

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PATH

12 Sophia observes a plane due north at an angle of elevation of 40°. The plane is flying due east at an altitude of 4000 m and at a speed of 200 km/h. PS R C

a Find the angle of elevation and the bearing of the plane from Sophia after 2 minutes.

b What is the distance (to the nearest m) from Sophia to the plane after 2 minutes?

13 From a lighthouse 95 m above sea level, a boat is observed on a compass bearing of SSE and at an angle of depression of 2°. Another boat is seen on a bearing of ENE and at angle of depression of 1.5°. PS R C

a Calculate the distance (correct to one decimal place) between the boats.

b What is the bearing of the first boat from the second boat?

Quiz
Mental skills 8

Mental skills 8: Maths without calculators ANSWERS ON P. 613

Divisibility tests

How can you tell if a number is divisible by 2? Look at its last digit. If that digit is 2, 4, 6, 8 or 0, then the number is divisible by 2 (that is, it is even).

How can you tell if a number is divisible by 5? If its last digit is 0 or 5, then the number is divisible by 5.

These are examples of **divisibility tests**—rules for checking whether or not a number is divisible by a certain number. The table below shows some common divisibility tests.

A number is divisible by:	if:
2	its last digit is 2, 4, 6, 8 or 0
3	the sum of its digits is divisible by 3
4	its last two digits form a number divisible by 4
5	its last digit is 0 or 5
6	it is even and the sum of its digits is divisible by 3
9	the sum of its digits is divisible by 9
10	its last digit is 0

1 Study each example.

a Test whether 748 is divisible by 2, 3 or 4.

- Last digit is 8 (even), ∴ 748 is divisible by 2
- Sum of digits = 7 + 4 + 8 = 19, which is not divisible by 3, ∴ 748 is not divisible by 3
- 48 is divisible by 4, ∴ 748 is divisible by 4 (748 ÷ 4 = 187)

b Test whether 261 is divisible by 5 or 9.

- Last digit is 1, not 0 or 5, ∴ 261 is not divisible by 5
- 2 + 6 + 1 = 9, which is divisible by 9, ∴ 261 is divisible by 9. (261 ÷ 9 = 29).

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c Test whether 570 is divisible by 4, 6 or 10.

- 70 is not divisible by 4, ∴ 570 is not divisible by 4
- 570 is even and 5 + 7 + 0 = 12, which is divisible by 3, ∴ 570 is divisible by 6 (570 ÷ 6 = 95)
- Last digit is 0, ∴ 570 is divisible by 10 (570 ÷ 10 = 57)

2 Test whether each number is divisible by 2, 3, 5 or 6.

a	250	**b**	189	**c**	78	**d**	465
e	1024	**f**	840	**g**	715	**h**	627

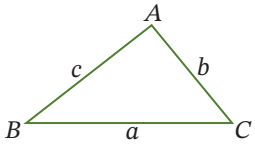
3 Test whether each number is divisible by 4, 9 or 10.

a	144	**b**	280	**c**	522	**d**	4170
e	936	**f**	726	**g**	342	**h**	5580

The sine rule

8.06

The angles of a triangle are labelled with capital letters while the sides are labelled with lowercase letters. By convention, we use a to label the side opposite $\angle A$, b to label the side opposite $\angle B$, and so on.



There is a relationship between each angle in a triangle and its opposite side. The longest side is always opposite the largest angle, the next smallest side is opposite the next smallest angle and so on. This relationship is called the **sine rule**.

PATH

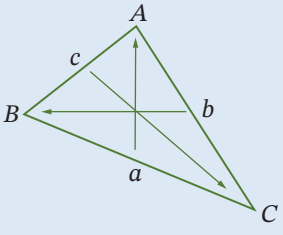
Worksheet
Discovering the sine rule

The sine rule

For any triangle ABC:

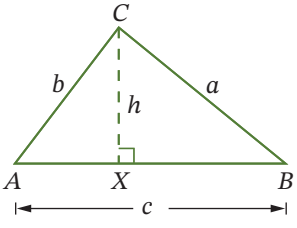
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

The ratios of the sides in a triangle to the sine of their opposite angles are equal.



Proof:

In $\triangle ABC$, draw CX for the perpendicular height, h, of the triangle. CX divides $\triangle ABC$ into 2 right-angled triangles.



In $\triangle AXC$, $\sin A = \frac{h}{b}$

$\therefore h = b \sin A$

In $\triangle BXC$, $\sin B = \frac{h}{a}$

$\therefore h = a \sin B$

$\therefore a \sin B = b \sin A$

PATH

$$\frac{a\sin B}{\sin B}=\frac{b\sin A}{\sin B}$$ Dividing both sides by $\sin B$

$$a=\frac{b\sin A}{\sin B}$$

$$\frac{a}{\sin A}=\frac{b\sin A}{\sin B\sin A}$$ Dividing both sides by $\sin A$

$$\frac{a}{\sin A}=\frac{b}{\sin B}$$

By drawing the perpendicular from A to BC, it can be shown that $\frac{b}{\sin B}=\frac{c}{\sin C}$

$$\therefore\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

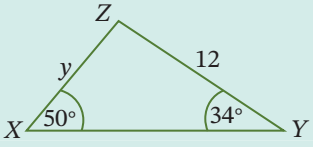
The sine rule allows us to apply trigonometry to any triangle, not just right-angled triangles.

The sine rule is used in problems involving 2 sides of a triangle and the 2 angles opposite them.

Video
The sine rule

Example 16

Find y, correct to one decimal place.



Solution

$$\frac{a}{\sin A}=\frac{b}{\sin B}$$ Sides and opposite angles

$$\frac{y}{\sin 34°}=\frac{12}{\sin 50°}$$

$$y=\frac{12\sin 34°}{\sin 50°}$$

$=8.7596\ldots$

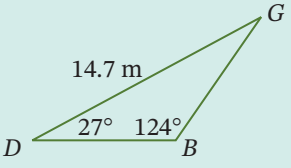
≈ 8.8 cm From the diagram, an answer of 8.8 cm looks reasonable

Example 17

In $\triangle BDG$, $\angle B=124°$, $\angle D=27°$ and $DG=14.7$ m. Find BD, correct to one decimal place.

Solution

Draw a diagram.



$$\frac{BD}{\sin G}=\frac{14.7}{\sin 124°}$$

$\angle G=180°-27°-124°$ (angle sum of a triangle)

$=29°$

$$\frac{BD}{\sin 29°}=\frac{14.7}{\sin 124°}$$

$$BD=\frac{14.7\sin 29°}{\sin 124°}$$

$=8.5963\ldots$

≈ 8.6 m

EXERCISE 8.06 ANSWERS ON P. 613

The sine rule U F PS R

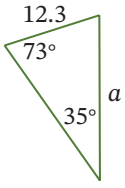
EXAMPLE 16

1 Evaluate each expression, correct to one decimal place.

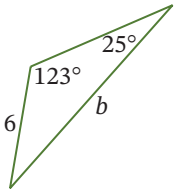
a $\dfrac{14.7\sin 64°}{\sin 46°}$ **b** $\dfrac{34.5\sin 33.4°}{\sin 115.7°}$ **c** $\dfrac{69\sin 107°\,33'}{\sin 38°\,47'}$

2 Find the value of each variable, correct to 2 decimal places.

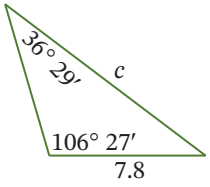
a



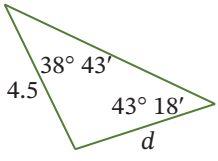
b



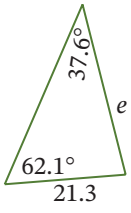
c



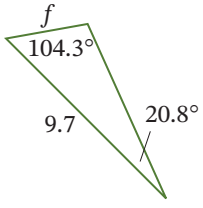
d



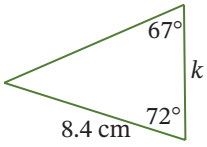
e



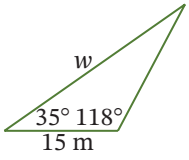
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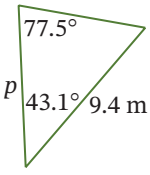
g



h



i

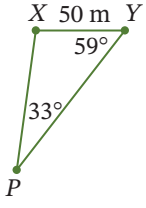


3 In $\triangle PTM$, $\angle P = 67°$, $\angle M = 52°$ and $PM = 12.5$ m. Find the length of PT.
Select the closest answer **A**, **B**, **C** or **D**.

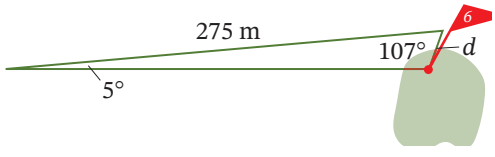
A 10.7 m **B** 14.6 m **C** 11.3 m **D** 13.9 m

EXAMPLE 17

4 X and Y are 2 light towers 50 m apart on one side of a park. P is a light tower on the other side of the park. If $\angle Y = 59°$ and $\angle P = 33°$, find PX correct to the nearest metre. PS R



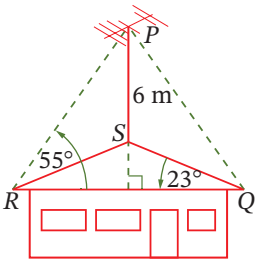
5 A golfer drives a ball 275 m at an angle of 5° off centre. The ball lands at an angle of 107° from the hole. Calculate the distance of the ball from the hole, correct to the nearest metre. PS R



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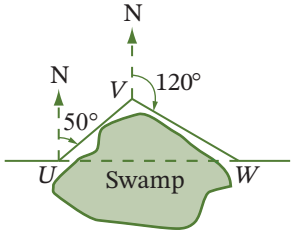
6 A 6 m television antenna is mounted on a roof pitched at an angle of 23°. It is supported by 2 wires, PQ and PR, inclined at 55° to the horizontal. PS R



a Show that $\angle PSR = 113°$.

b Calculate the length of the wire PR, correct to the nearest centimetre.

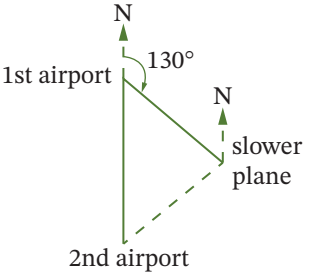
7 To avoid a swamp, Jesinta runs 70 m on a bearing of 050° to V. She then turns and runs to W on a bearing of 120°. If W is directly east of U: PS R



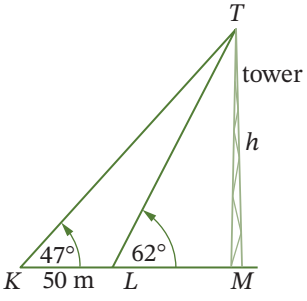
a find $\angle UVW$

b calculate UW, correct to one decimal place.

8 2 planes leave the airport at the same time. One flies due south at 400 km/h and lands at a second airport after $1\frac{1}{2}$ hours. The other flies on a bearing of S50°E and after $1\frac{1}{2}$ hours is at a bearing of N75°E from the second airport. How far (to the nearest km) is the slower plane from the second airport? PS R



9 The angle of elevation of a tower from a point L is 62°. From a point K, 50 m further from the tower, the angle of elevation is 47°. PS R



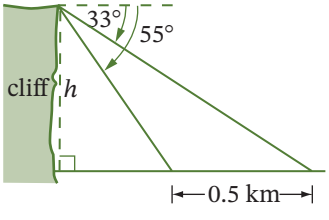
a Use the sine rule in $\triangle KTL$ to show that $TL = \frac{50 \sin 47°}{\sin 15°}$.

b Let the height of the tower be h. In the right-angled $\triangle LMT$, show that $TL = \frac{h}{\sin 62°}$.

c Hence show that $h = \frac{50 \sin 47° \sin 62°}{\sin 15°}$

d Hence calculate the height of the tower, correct to one decimal place.

10 From the top of a cliff, the angles of depression of 2 boats at sea that are 0.5 km apart are 55° and 33°. PS R



a Let the height of the cliff be h. Show that $h = \frac{0.5 \sin 33° \sin 55°}{\sin 22°}$.

b Hence calculate the height, to the nearest metre.

11 A yacht leaves Fremantle and sails on a bearing of 312° for 650 km. It then changes direction and sails on a bearing of 235° until it is due west of Fremantle. How far is the yacht from Fremantle (to the nearest km)? PS R

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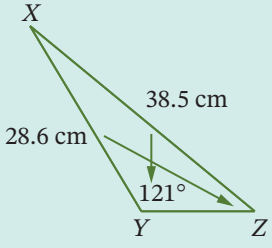
Hilke Maunder/Alamy Stock Photo

The sine rule for angles

8.07

Example 18

Find angle Z, correct to the nearest degree.



Solution

$$\frac{28.6}{\sin Z} = \frac{38.5}{\sin 121°}$$

Sides and opposite angles

$$\frac{\sin Z}{28.6} = \frac{\sin 121°}{38.5}$$

Inverting both sides so that Z is in the numerator

$$\sin Z = \frac{28.6 \sin 121°}{38.5}$$
$$= 0.636\ldots$$
$$Z = 39.55...$$
$$\approx 40°$$

On a calculator: SHIFT sin Ans =

From the diagram, an answer of 40° looks reasonable

PATH

Worksheet
Sine rule problems

Video
Obtuse angles using the sine rule

PATH

Obtuse-angled answers

If sin θ is positive, angle θ could be 2 different values, one acute and one obtuse.

However, these 2 values are related: they are supplementary angles (adding to 180°).

For example, if sin θ = 0.5, then θ = 30° or 180° – 30° = 150°.

You can check on your calculator that sin 30° = sin 150° = 0.5.

When using SHIFT sin on your calculator, it will give the acute angle θ only.

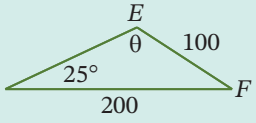
For the obtuse angle, you must calculate 180° – θ.

The sine ratio of supplementary angles

$\sin(180° - \theta) = \sin\theta$

Example 19

Find θ correct to the nearest minute if it is an obtuse angle.



Solution

$$\frac{200}{\sin\theta} = \frac{100}{\sin 25°}$$

$$\frac{\sin\theta}{200} = \frac{\sin 25°}{100}$$

$$\sin\theta = \frac{200\ \sin 25°}{100}$$

$$= 0.845\ldots$$

$\theta = 57.697...$

But θ is obtuse, so:

$\theta = 180 - 57.697...$

$= 122.3027...$

$= 122°\ 18'\ 9.77''$

$\approx 122°\ 18'$

The ambiguous case (when there are 2 possible answers)

When we use the sine rule to find an angle, it is possible to find both an acute angle and an obtuse angle as solutions. Likewise, there could be 2 possible triangles: one acute-angled, the other obtuse-angled. However, the obtuse-angled triangle may not be possible. We need to check that the sum of the angles in the triangle is not greater than 180°.

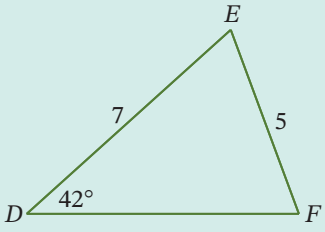
Example 20

a In $\triangle DEF$, $\angle D = 42°$, $d = 5$ cm and $f = 7$ cm. Find $\angle F$ correct to the nearest degree.

b In $\triangle LMN$, $\angle M = 130°$, $LN = 15$ cm and $LM = 7$ cm. Find $\angle N$, correct to the nearest degree.

Solution

a Draw a rough diagram.



$$\frac{7}{\sin F} = \frac{5}{\sin 42°}$$

$$\frac{\sin F}{7} = \frac{\sin 42°}{5}$$

$$\sin F = \frac{7 \sin 42°}{5}$$

$$= 0.93678\ldots$$

$$F = 69.5181...$$

$$\approx 70°$$

But F could be obtuse.

$$F = 180° - 70°$$

$$= 110°$$

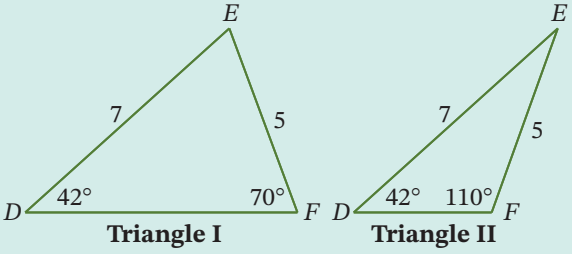
Checking the third angle of the obtuse-angled triangle:

$$\angle E = 180° - 42° - 110°$$

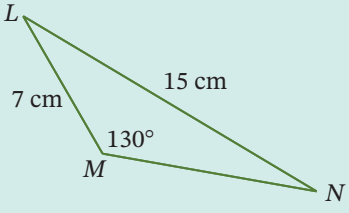
$$= 28°$$

$\therefore$ The obtuse-angled solution is possible.

$\therefore \angle F = 70°$ or $110°$



b Draw a rough diagram.



$$\frac{\sin N}{7} = \frac{\sin 130°}{15}$$

$$\sin N = \frac{7 \sin 130°}{15}$$

$$= 0.3574...$$

$$N = 20.9459...$$

$$\approx 21°$$

PATH

Video
The sine rule

8.07

PATH

But N could be obtuse.

$N = 180° - 21°$

$= 159°$

Checking the third angle of the obtuse-angled triangle:

$\angle L = 180° - 130° - 159°$ Impossible

$= -109°$

$\therefore$ The obtuse-angled solution is not possible.

$\therefore \angle N = 21°$

EXERCISE 8.07 ANSWERS ON P. 613

The sine rule for angles U F PS R C

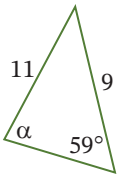
1 Find the acute angle X in each equation, correct to the nearest degree.

a $\sin X = \frac{5.3 \sin 123°}{9.7}$ **b** $\sin X = \frac{39 \sin 85°29'}{64}$ **c** $\sin X = \frac{467 \sin 63.8°}{518}$

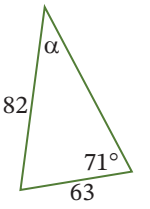
EXAMPLE 18

2 Find α in each triangle if α is acute, correct to the nearest 0.1 degree.

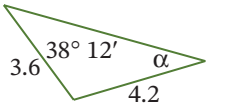
a



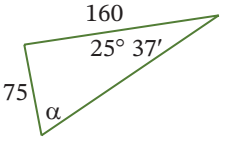
b



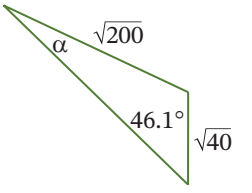
c



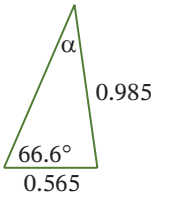
d



e



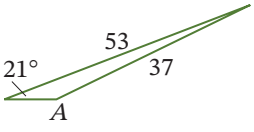
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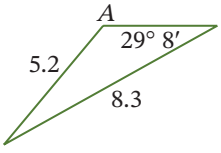
EXAMPLE 19

3 Find the size of $\angle A$ to the nearest minute if $\angle A$ is obtuse.

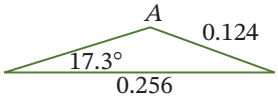
a



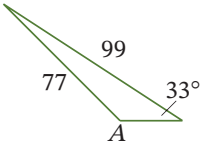
b



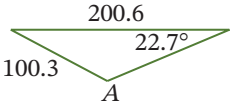
c



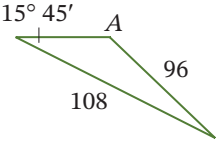
d



e



f

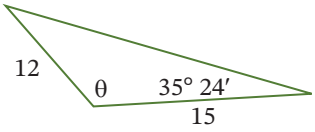


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PATH

4 Find the size of θ. Select the correct answer **A**, **B**, **C** or **D**.



A 46° 24′ **B** 98° 12′

C 133° 36′ **D** 81° 48′

5 In $\triangle BGP$, $\angle G = 121°$, $BG = 5.2$ m, and $BP = 12.8$ m. Find $\angle B$ (to the nearest degree).

6 Find all possible angles for each triangle, correct to the nearest degree. R C

EXAMPLE 20

a In $\triangle PQR$, $\angle P = 35°$, $p = 8$ cm, and $q = 10$ cm. Find $\angle Q$.

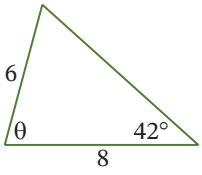
b In $\triangle UVW$, $\angle W = 95°$, $w = 16$ cm, and $v = 10$ cm. Find $\angle V$.

c In $\triangle XYZ$, $\angle Y = 24°$, $y = 3.4$ km, and $z = 5.7$ km. Find $\angle Z$.

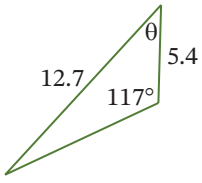
d In $\triangle DEF$, $\angle E = 37°$, $e = 107$ mm, and $d = 121$ mm. Find $\angle D$.

7 Find θ in each triangle correct to the nearest degree, given that θ is acute.

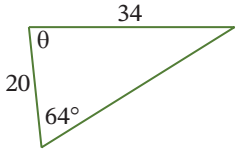
a



b



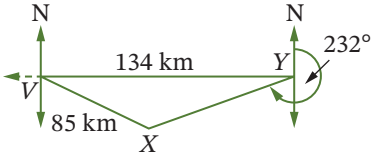
c



8 Max walked 5.6 km due east from his campsite. He then turned and walked on a bearing of S33°W until he was 6 km from the campsite. What is the bearing of the campsite from Max? Select **A**, **B**, **C** or **D**. PS R C

A N52°E **B** N71°W **C** N19°W **D** N52°W

9 Yorker (Y) is 134 km due east of Vendor (V). Xena (X) is on a bearing of 232° from Yorker and is also 85 km from Vendor. $\angle VXY$ is an obtuse angle. Find the bearing of Xena from Vendor. PS R C



8.08

The cosine rule

8.08

PATH

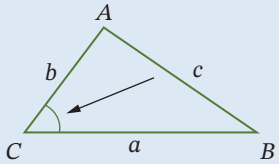
The **cosine rule** is a relationship between the 3 sides of a triangle and one of its angles. It is an extension of Pythagoras' triangle that can be applied to any triangle, not just right-angled ones.

The cosine rule

For any triangle ABC:

$$c^2 = a^2 + b^2 - 2ab\cos C$$

where c is the unknown side, C is the angle opposite c, and a and b are the other 2 sides.

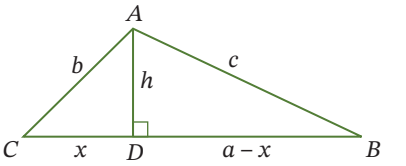


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Proof:

In $\triangle ABC$, draw AD for the perpendicular height, h, of the triangle. AD divides $\triangle ABC$ into 2 right-angled triangles.



Let $CD = x$, $\therefore DB = a - x$

In $\triangle ADC$, $b^2 = h^2 + x^2$

$\therefore h^2 = b^2 - x^2$ [1]

In $\triangle ABD$, $c^2 = h^2 + (a - x)^2$

$\therefore h^2 = c^2 - (a - x)^2$

$\therefore b^2 - x^2 = c^2 - (a - x)^2$

$b^2 - x^2 = c^2 - (a^2 - 2ax + x^2)$

$b^2 - x^2 = c^2 - a^2 + 2ax - x^2$

$b^2 = c^2 - a^2 + 2ax$

$\therefore c^2 = a^2 + b^2 - 2ax$ [2]

In $\triangle ADC$, $\cos C = \frac{x}{b}$

$\therefore x = b \cos C$

Substituting for x in [2]:

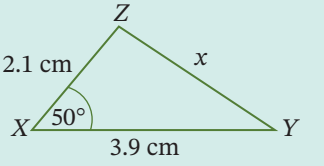
$c^2 = a^2 + b^2 - 2ab \cos C$

The cosine rule can be used in problems involving 3 sides of a triangle and one of the angles.

Video
The cosine rule

Example 21

Find x correct to 2 decimal places.



Solution

$c^2 = a^2 + b^2 - 2ab \cos C$

$x^2 = 2.1^2 + 3.9^2 - 2 \times 2.1 \times 3.9 \cos 50°$ — 50° is the angle opposite x

$= 9.091138...$

$x = \sqrt{9.091138...}$

$= 3.01515...$

≈ 3.02 cm — From the diagram, an answer of 3.02 cm looks reasonable

EXERCISE 8.08 ANSWERS ON P. 613

The cosine rule U F PS R C

1 Solve each equation for x, correct to one decimal place.

a $x^2 = 8^2 + 9^2 - 2 \times 8 \times 9 \times \cos 38°$

b $x^2 = 11.3^2 + 9.7^2 - 2 \times 11.3 \times 9.7 \times \cos 76.9$

c $x^2 = 17^2 + 20.1^2 - 2 \times 17 \times 20.1 \times \cos 149° 45'$

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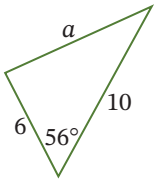
PATH

EXAMPLE 21

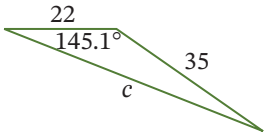
8.08

2 Find, correct to 2 decimal places, the value of each variable.

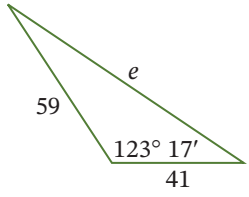
a



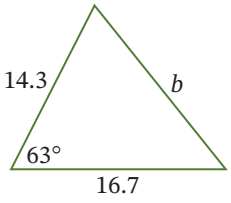
b



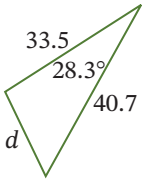
c



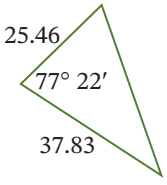
d



e



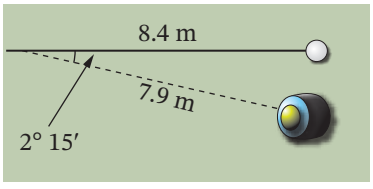
f



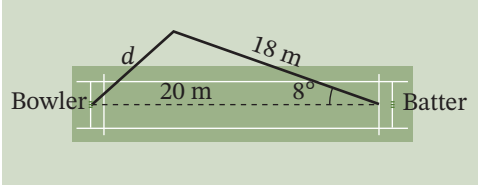
3 In $\triangle DEF$, $\angle E = 38° 46'$, $d = 8.7$ m and $f = 15.6$ m. What is the length of DF?
Select the closest answer **A**, **B**, **C** or **D**.

A 10.4 m **B** 19.5 m **C** 17.9 m **D** 12.8 m

4 In a game of lawn bowls, Jayden is aiming to hit the jack (target ball) 8.4 m away. If he bowls 2° 15′ off centre and his bowl travels 7.9 m, how far is his bowl from the jack? Answer correct to one decimal place. PS R

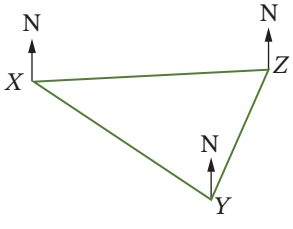


5 In a cricket match, the distance between the bowler and the batter was 20 m. During one bowl, the batter hit the ball at an angle of 8° to the line of the pitch and the bowler ran and caught the ball after it had travelled 18 m. How far did the bowler run to catch the ball?
Select **A**, **B**, **C** or **D**. PS R



A 1.1 m **B** 2.0 m **C** 3.3 m **D** 4.0 m

6 A yacht sails from X to Y on a bearing of 130° for 4.2 km. It then turns and travels to Z on a bearing of 025° for 2.9 km. PS R C



a Copy the diagram and mark the given information on it.

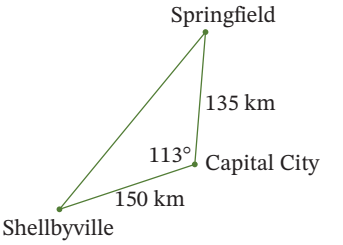
b Explain why $\angle XYZ = 75°$.

c Calculate the distance XZ, correct to one decimal place.

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7 3 towns are joined by straight roads. What distance (correct to the nearest kilometre) is saved by going directly from Springfield to Shellbyville instead of travelling via Capital City? PS R C



8 **a** What is the value of cos 90°?

b What does $c^2 = a^2 + b^2 - 2ab\cos C$ simplify to if $C = 90°$?

c Hence what happens to the cosine rule when it is applied to a right-angled triangle?

8.09 The cosine rule for angles

PATH

Worksheet
Cosine rule problems

If we rewrite the cosine rule so that cos A is the subject, then we will have a formula for finding an unknown angle when the 3 sides of a triangle are known.

$$c^2 = a^2 + b^2 - 2ab\cos C$$

$c^2 + 2ab\cos C = a^2 + b^2$ Adding $2ab\cos C$ to both sides so that cos C appears on the LHS

$$2ab\cos C = a^2 + b^2 - c^2$$

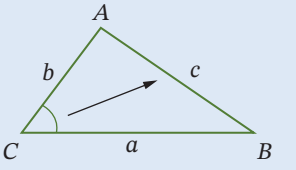
$$\therefore \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

The cosine rule for angles

For any triangle ABC:

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

where C is the unknown angle, c is the side opposite C, and a and b are the other 2 sides.

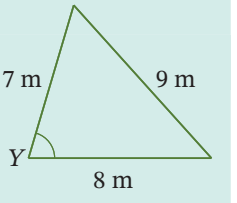


The cosine rule can be used to find an unknown angle if the lengths of the 3 sides are known.

Example 22

Video
The cosine rule for angles 2

Find the size of the marked angle Y, correct to the nearest degree.



Solution

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos Y = \frac{7^2 + 8^2 - 9^2}{2 \times 7 \times 8}$$

$$= \frac{32}{112}$$

9 mm is opposite angle Y

$Y = 73.398...$

$\approx 73°$

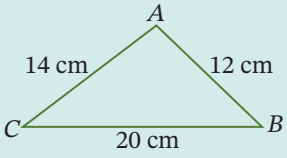
From the diagram, an answer of 73° looks reasonable

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Example 23

Calculate, correct to the nearest minute, the size of the largest angle in this triangle.



Solution

The largest angle is opposite the longest side, so it is $\angle A$.

$$\cos A = \frac{14^2 + 12^2 - 20^2}{2 \times 14 \times 12}$$

$$= \frac{-60}{336}$$

Cos is negative so the angle will be obtuse

$$A = 100.28656...$$

$$= 100° \, 17' \, 11.6''$$

$$\approx 100° \, 17'$$

From the diagram, an answer of 100° 17′ looks reasonable

PATH

Videos
The cosine rule
The cosine rule for angles 1

8.09

EXERCISE 8.09 ANSWERS ON P. 613

The cosine rule for angles U F PS R

1 Solve each equation for X, correct to the nearest degree.

a $\cos X = \frac{12^2 + 14^2 - 15^2}{2 \times 12 \times 14}$

b $\cos X = \frac{5.7^2 + 6.8^2 - 3.7^2}{2 \times 5.7 \times 6.8}$

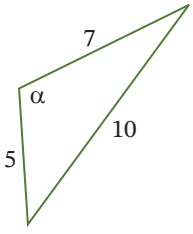
c $\cos X = \frac{5^2 + 6^2 - 9^2}{2 \times 5 \times 6}$

d $\cos X = \frac{9.2^2 + 4.7^2 - 12.8^2}{2 \times 9.2 \times 4.7}$

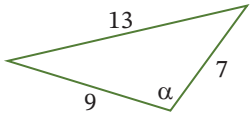
EXAMPLE 22

2 Find the size of α in each triangle, correct to the nearest degree.

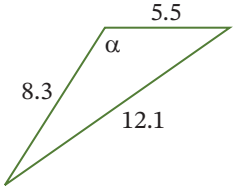
a



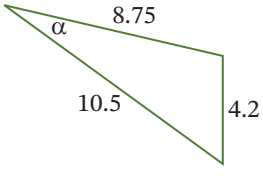
b



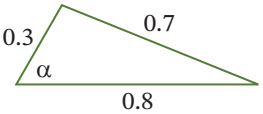
c



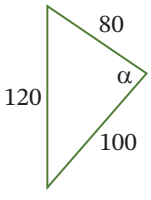
d



e



f



3 In $\triangle ABC$, $a = 8.5$ cm, $b = 12.1$ cm and $c = 17.6$ cm. What is the size of $\angle B$?
Select the correct answer **A**, **B**, **C** or **D**.

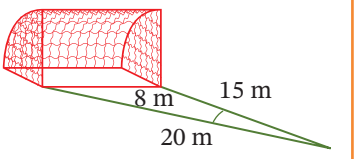
A 25° 40′ **B** 141° 57′ **C** 116° 17′ **D** 38° 3′

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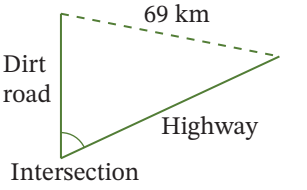
PATH

EXAMPLE 23

4 A soccer goal is 8 m wide. A player shoots for goal (along the ground) when 20 m from one post and 15 m from the other post. Within what angle (correct to the nearest 0.1 degree) must the shot be made for the player to have a chance of scoring a goal? PS R



5 2 cars leave an intersection at the same time. Car *A* drives down the dirt road at 60 km/h and car *B* drives down the highway at 100 km/h. After 45 minutes they are 69 km apart. Find the angle between the 2 roads, correct to the nearest minute. PS R



6 A triangle has sides of 21 m, 17 m and 10 m.
Find the size of the largest angle, correct to the nearest degree. R

7 A plane flies from Alice Springs on a bearing of S65°W. After 450 km, the plane changes to a northwesterly direction for 1100 km. When the plane stops it is 1380 km from Alice Springs. What is the bearing (to the nearest degree) of the plane from Alice Springs? PS R

8.10 The area of a triangle

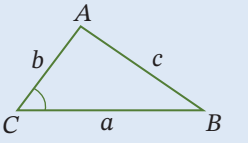
PATH

We already know that the formula for the area of a triangle is $A = \frac{1}{2}bh$, but there is also a trigonometric formula if we know the lengths of 2 sides of the triangle and the size of the **included angle** between them.

The area of a triangle

$A = \frac{1}{2}ab\sin C$

where C is the included angle between sides a and b.



Proof:

In $\triangle ABC$, draw AX for the perpendicular height, h, of the triangle.

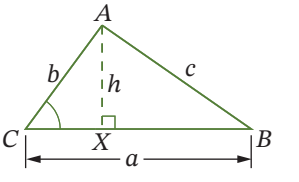
Area of $\triangle ABC = \frac{1}{2}ah$ [1]

But in $\triangle AXC$, $\sin C = \frac{h}{b}$

$\therefore h = b \sin C$

Substituting this into [1]:

$A = \frac{1}{2}ab\sin C$

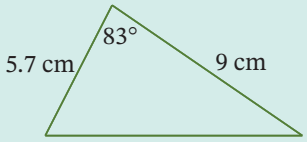


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Example 24

Find, correct to one decimal place, the area of this triangle.



Solution

$A = \frac{1}{2}ab\sin C$

$= \frac{1}{2} \times 5.7 \times 9 \times \sin 83°$

$= 25.458...$

$\approx 25.5 \text{ cm}^2$

83° is the included angle between 5.7 cm and 9 cm

PATH

Video
The sine area formula

8.10

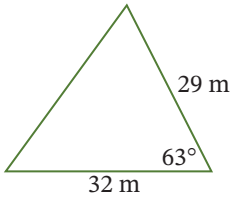
EXAMPLE 24

EXERCISE 8.10 ANSWERS ON P. 613

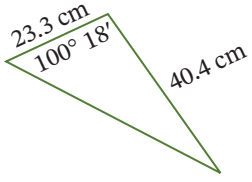
The area of a triangle U F PS R

1 Find, correct to one decimal place, the area of each triangle.

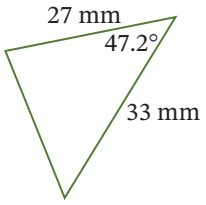
a



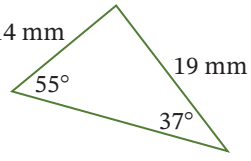
b



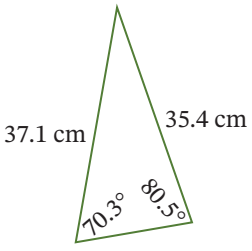
c



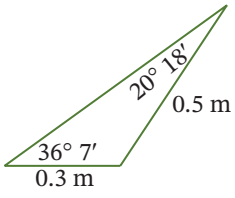
d



e

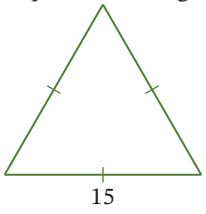


f

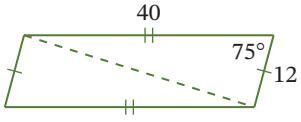


2 Calculate, correct to one decimal place, the area of each shape. All measurements are in metres. R

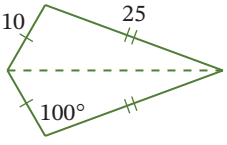
a Equilateral triangle



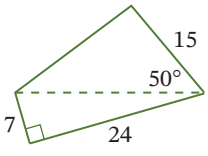
b Parallelogram



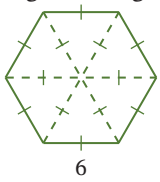
c Kite



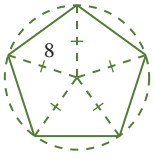
d Quadrilateral



e Regular hexagon



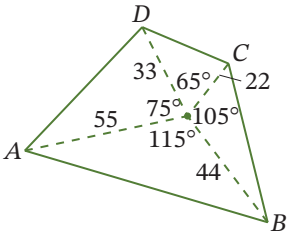
f Regular pentagon inscribed in a circle of radius 8



Foundation | Standard | Complex

PATH

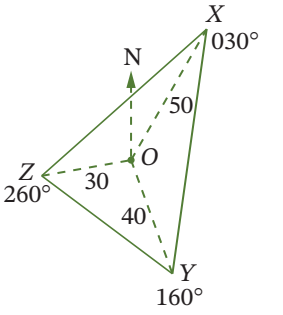
3 The diagram shows the results of a radial survey of a block of land. All distances are in metres. PS R



a Use the cosine rule to find the lengths of AB, BC, CD and AD and, hence, find the perimeter of the block of land (to the nearest metre).

b Use the area formula to find the area of each triangle and, hence, calculate the area of the block of land (to the nearest m^2).

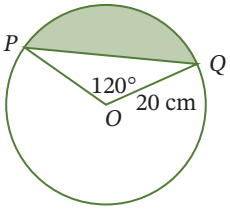
4 The results of a radial survey are shown in the diagram. All measurements are in metres. PS R



a Find the size of $\angle XOY$.

b Calculate, correct to 2 decimal places, the area of $\triangle XOY$.

5 O is the centre of a circle of radius 20 cm. Calculate, correct to one decimal place, the area of: PS R

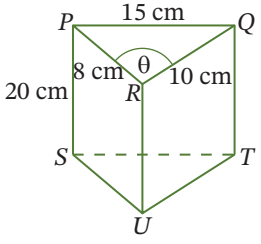


a sector OPQ

b triangle OPQ

c the shaded segment.

6 A triangular prism has base edges of 8 cm, 10 cm and 15 cm, and a height of 20 cm. PS R



a Calculate the size $\angle PRQ$, correct to nearest degree.

b Find the area of $\triangle PQR$, correct to the nearest cm^2.

c Find the volume of the prism.

7 The area of an equilateral triangle is 600 cm^2. Calculate the perimeter of the triangle, correct to one decimal place.

Did you know?

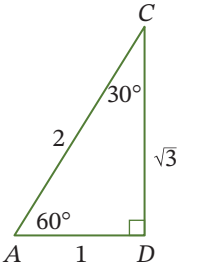
Triangle area formulas

There are many formulas for calculating the area, A, of a triangle:

Heron's formula: given 3 sides a, b, c:

$A = \sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter $\frac{a+b+c}{2}$

- Given 2 sides, a, b and the included angle, C: $A = \frac{1}{2}ab \sin C$
- Given one side, a and 3 angles, A, B, C: $A = \frac{1}{2}a^2 \frac{\sin B \sin C}{\sin A}$



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1 **Find the area of this triangle using the 3 formulas, and check that your answers agree.**

2 **Show how the formula $A = \frac{1}{2}a^2 \frac{\sin B \sin C}{\sin A}$ can be proved from $A = \frac{1}{2}ab \sin C$ using the sine rule.**

3 **Research other formulas for the area of a triangle.**

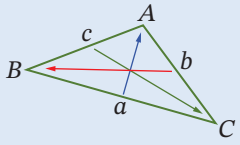
Problems involving the sine and cosine rules

8.11

The sine and cosine rules

The sine rule is used for triangle problems involving 2 sides and the 2 angles opposite them.

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

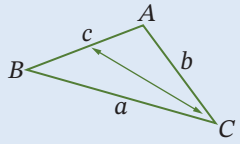


The cosine rule is used for triangle problems involving 3 sides and one angle.

$$c^2 = a^2 + b^2 - 2ab \cos C$$

and

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



PATH

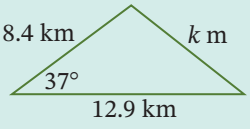
Worksheets
Finding an unknown side
Finding an unknown angle
The sine and cosine rules

Puzzle
The sine and cosine rules

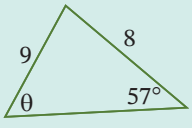
Video
The sine and cosine rules

Example 25

a Find the value of k, correct to one decimal place.



b Find the value of θ, correct to the nearest minute.



Solution

a The problem involves 3 sides and one angle so use the cosine rule.

$k^2 = 8.4^2 + 12.9^2 - 2 \times 8.4 \times 12.9 \times \cos 37°$

$= 63.889...$

$k = \sqrt{63.889...}$

$= 7.993...$

≈ 8.0 m

b The problem involves 2 sides and the 2 angles opposite them, so use the sine rule.

$\frac{\sin \theta}{8} = \frac{\sin 57°}{9}$

$\sin \theta = \frac{8 \sin 57°}{9}$

$= 0.7454...$

$\theta = 48.2007...$

$= 48° 12' 2.77''$

$\approx 48° 12'$

PATH

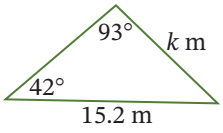
EXERCISE 8.11 ANSWERS ON P. 613

Problems involving the sine and cosine rules U F PS R C

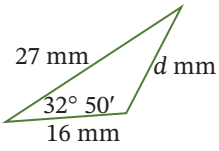
EXAMPLE 25

1 Find, correct to one decimal place, the value of each variable.

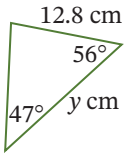
a



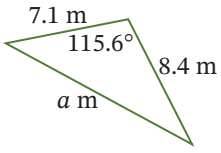
b



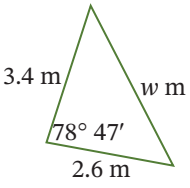
c



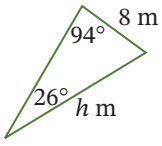
d



e

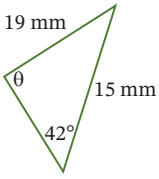


f

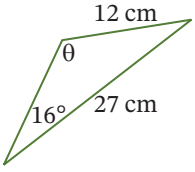


2 Find the value of θ to the nearest degree. Use diagrams to note whether θ is acute or obtuse.

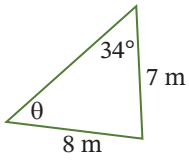
a



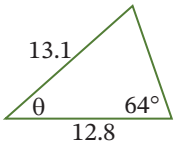
b



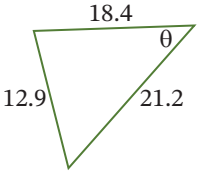
c



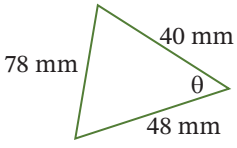
d



e



f

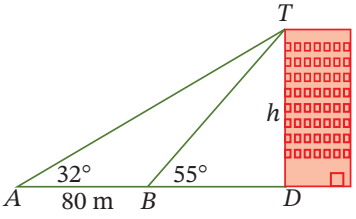


3 The angles of elevation of a building measured from 2 positions 80 m apart are 32° and 55°. PS R C

a Explain why $\angle ATB = 23°$.

b Find, correct to 2 decimal places, the length of BT.

c Hence find the height, h, of the building, correct to the nearest metre.



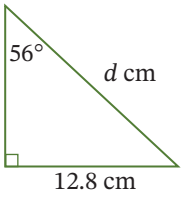
4 **a** What is the value of sin 90°? PS R C

b Find, correct to one decimal place, the value of d using:

i the sine rule

ii the sine ratio for right-angled triangles.

c What do you notice about your results? Give reasons.



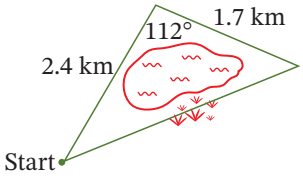
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PATH

8.11

5 Mikayla needs to run around a cross-country course as shown. What is the length of the course, correct to one decimal place? PS R



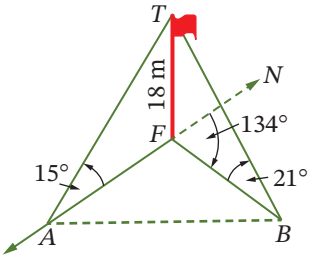
6 A plane flew on a bearing of 150° for 370 km. It then changed direction and flew another 285 km on a bearing of 235°. How far, correct to the nearest km, is the plane from its starting point? PS R

7 From a point A, south of a flagpole, the angle of elevation to the top of the pole is 15°. From B, the angle of elevation to the top of the flagpole is 21°.

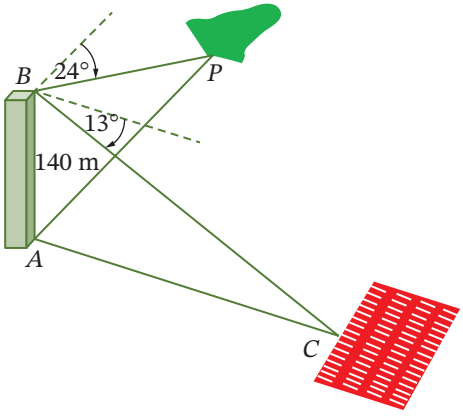
The bearing of B from the flagpole is S46°E. Find correct to one decimal place: PS R

a the length of AF and BF

b the distance between A and B



8 From the top of a building, the angles of depression to a park (P) and a car park (C) are 24° and 13° respectively. The height of the building is 140 m, the park is due north of the building and the bearing of the car park is ESE from the building. Calculate the distance between the park and the car park, correct to the nearest metre. PS R



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Power plus ANSWERS ON P. 614

1 **a** Copy and complete each pair of trigonometric ratios, correct to 3 decimal places.

i $\sin 20° =$ _____, $\cos 70° =$ _____

ii $\sin 47° =$ _____, $\cos 43° =$ _____

iii $\sin 55° =$ _____, $\cos 35° =$ _____

iv $\sin 85° =$ _____, $\cos 5° =$ _____

b What do you notice about each pair of answers in part **a**?

c What do you notice about each pair of angles in part **a**?

d If $\cos 30° \approx 0.8660$ and $\sin \theta \approx 0.8660$, what is the value of θ?

e Copy and complete each equation.

i $\sin 75° = \cos$ ____ **ii** ____ $80° = \cos 10°$

iii $\cos$ ____ $= \sin 72°$ **iv** $\sin 30° =$ ____ $60°$

v $\cos 65° = \sin$ ____ **vi** $\sin$ ____ $= \cos 58°$

f Copy and complete this general rule: $\sin x = \cos$ (_____).

g Use a right-angled triangle with one angle x and sides a, b and c to prove that the above rule is true.

2 A plane is flying at an angle of 15° inclined to the horizontal.

a How far will the plane have to travel along its line of flight to increase its altitude (height) by 500 m?

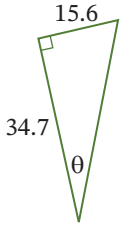
b At what angle must the plane climb to achieve an increase in altitude of 500 m in half the distance needed at an angle of 15°?

3 If $\sin 30° = \frac{1}{2}$, express the following as surds.

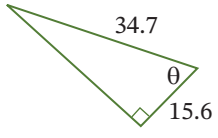
a $\cos 30°$ **b** $\tan 30°$

4 Find the value of angle θ, correct to the nearest second.

a



b



5 By drawing an appropriate triangle, prove that:

a $\tan 45° = 1$ **b** $\sin 45° = \frac{1}{\sqrt{2}}$ **c** $\cos 45° = \frac{1}{\sqrt{2}}$

6 Sketch a right-angled triangle with side lengths a, b and c, where c is the hypotenuse.

Let θ be one of the acute angles. Prove that $(\sin \theta)^2 + (\cos \theta)^2 = 1$.

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CHAPTER 8 REVIEW

Quiz
Language of maths 8

Puzzle
Further trigonometry crossword

Language of maths

adjacent	angle of depression	angle of elevation	bearing
clinometer	compass bearing	cosine rule	degree (°)
denominator	horizontal	hypotenuse	included angle
inverse ($^{-1}$)	minute (′)	obtuse	opposite
right-angled	sine rule	supplementary	tangent (tan)
theta (θ)	trigonometric ratio	true bearing	vertical

1 Copy and complete: A **bearing** is an _____ used to describe precisely the _____ of one location from a given reference point.

2 The word **minute** has an alternative pronunciation and meaning. What is its alternative meaning?

3 If 2 angles have a sum of 180°, what special name do we give them?

4 Explain what is meant by an **angle of depression**.

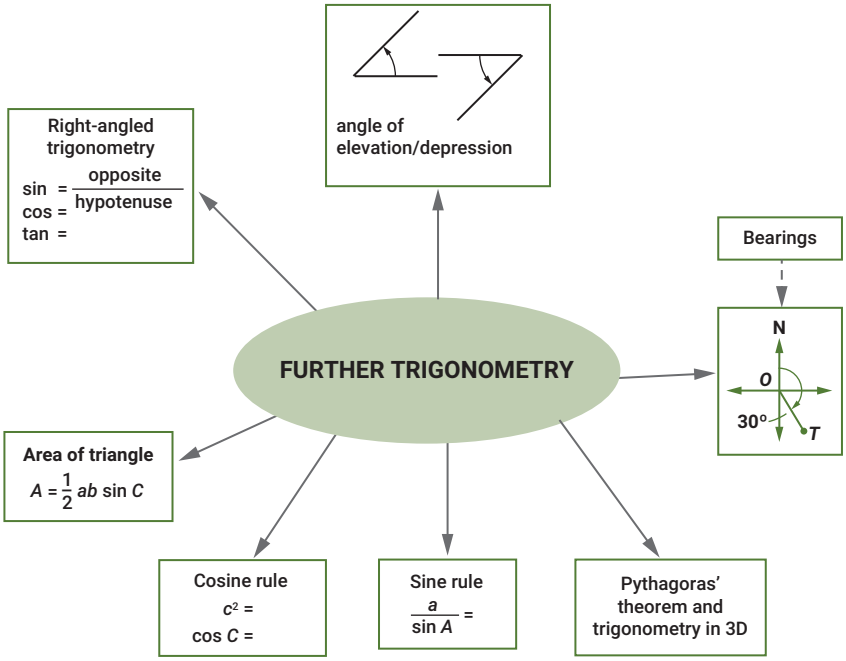
5 What type of bearing is also called a **three-figure bearing**?

6 What is the name given to the angle that is between 2 known sides?

Topic summary

Worksheet
Mind map: Further trigonometry

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.

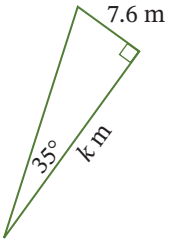


Quiz
Test yourself 8

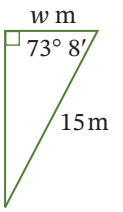
TEST YOURSELF 8 ANSWERS ON P. 614

8.01 **1** Find, correct to one decimal place, the value of each variable.

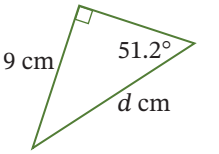
a



b

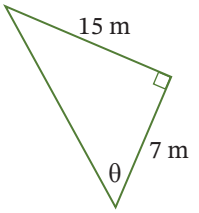


c

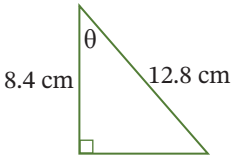


8.01 **2** Find the value of θ, correct to the nearest minute.

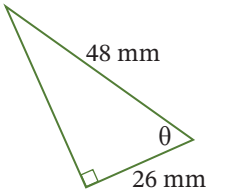
a



b



c



8.01 **3** In $\triangle XWY$, $\angle W = 90°$, $WY = 17.2$ m and $XW = 3.5$ m. Find $\angle Y$, correct to the nearest minute. Select the correct answer **A**, **B**, **C** or **D**.

A 11° 30′ **B** 11° 50′ **C** 11.52′ **D** 11° 31′

8.02 **4** From the top of a 300 m cliff Renee observes a boat at a distance of 1.3 km from the base of the cliff. Find, correct to the nearest degree, the angle of depression of the boat.

8.02 **5** The angle of elevation of a tower roof is 26° at a point 400 m from its base. Find the height of the tower, correct to the nearest metre.

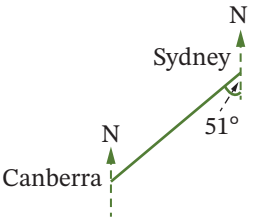
8.03 **6** Sketch point B on a compass rose so that B has a true bearing from A of:

a 220° **b** 020° **c** 120° **d** 310°

8.03 **7** What is the true bearing of:

a Canberra from Sydney?

b Sydney from Canberra?



8.04 **8** A hiking group walked on a bearing of N65°W for 13 km to reach a lookout.

a Find correct to one decimal place how far west they have travelled.

b Find the true bearing the group must walk to return to their starting point.

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9 2 planes leave an airport at the same time. The first travels on a bearing of 206° at 450 km/h. The second travels on a bearing of 296° at 600 km/h.

a How far does each plane travel after 3 hours?

b How far apart are the planes after 3 hours, to the nearest km?

c Calculate, correct to the nearest degree, the bearing of the first plane from the second plane.

8.04

10 A box in the shape of a square prism has a base of 10 cm by 10 cm and is 30 cm tall. Find to the nearest whole number:

a the length of the longest diagonal of the box

b the angle that the longest diagonal makes with the base.

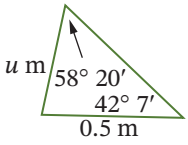
PATH

8.05

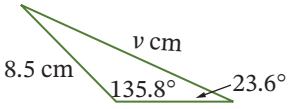
11 Find the value of each variable, correct to one decimal place.

8.06

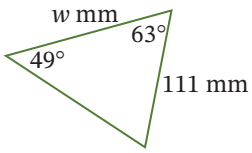
a



b



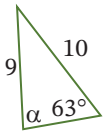
c



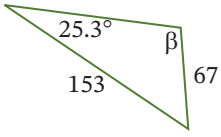
12 Find the value of each variable, correct to the nearest minute.

8.07

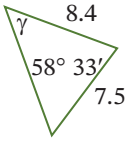
a



b



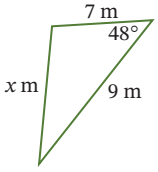
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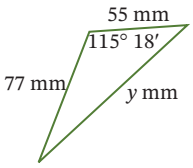
13 Find the value of each variable, correct to one decimal place.

8.08

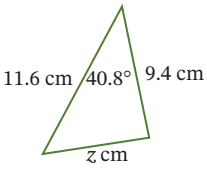
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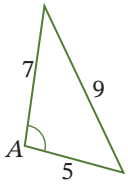
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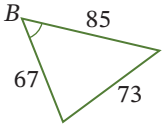
14 Find the size of each marked angle, correct to the nearest degree.

8.09

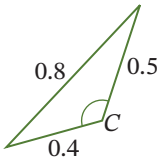
a



b



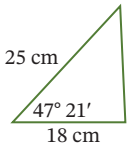
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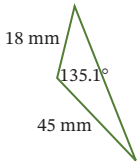
15 Find, correct to the nearest whole number, the area of each triangle.

8.10

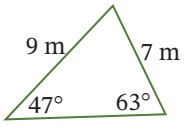
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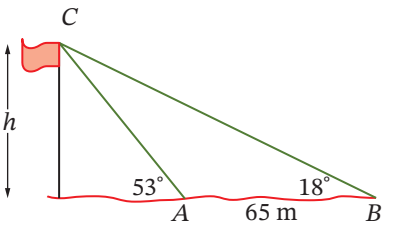
TEST YOURSELF 8

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8.11 **16** A parallelogram has sides of 12 cm and 6 cm and one interior angle of 65° 35′. Find the length of the longer diagonal, correct to one decimal place.

8.11 **17** The distance from A to B is 65 m.



a Show that $\angle ACB = 35°$

b Hence find the length of AC correct to 2 decimal places.

c Hence find the height of the flagpole correct to one decimal place.

8.11 **18** A boat sails from Perth on a bearing of SSW for 12 nautical miles and then changes to a bearing of WNW and sails a further 42 nautical miles. Find correct to one decimal place:

a how far the boat is from Perth

b the true bearing of the boat from Perth

c on what true bearing it will have to sail on its way directly back to Perth